The Children of Welfare Reform: Early Childhood Under TANF and Adult Outcomes*

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Abstract

In models of child development, a child's later outcomes depend partly on family income and parental time. Welfare rules that require work trade one for the other, so theory does not say whether children gain or lose. I study the 1990s welfare reforms that imposed such rules, following young children into adulthood in the Panel Study of Income Dynamics. I compare children born in the same year across states whose rules differed in strictness, by their families' likelihood of receiving welfare. Among children the rules likely reached, those with more exposure to stricter rules before age five had better general and mental health as adults. But their adult employment did not change, and the estimates for their hours and earnings are imprecise. On net, my findings suggest that these children gained as adults from the trade-off between family income and parental time that the rules imposed.

JEL codes: I38, I14, J13, H53

1 Introduction

In economic models of child development, a young child's later outcomes depend partly on family income and parental time (Del Boca et al., 2014; Bernal and Keane, 2011; Mullins, 2026). Parental employment raises income and costs time, so theory does not say whether a parent's work leaves a young child better or worse off (Del Boca et al., 2014). When a welfare rule requires work, every parent it covers must make that choice. A parent who complies gains earnings but spends less time at home. A parent who does not comply keeps that

*Latest version: `https://m-baratian.github.io/Baratian_JMP.pdf`

time but loses the aid. The welfare reform of the 1990s tied cash aid for low-income families to work. This paper measures the net effect on the young children in those families of the income gained and the time lost. It follows those children into adulthood and asks whether more years under stricter rules changed their health and their employment as adults.

The evidence so far points one way for income and both ways for time. An extra $1,000 of annual family income (in 2000 dollars) raises children's cognitive scores by about 3 percent of a standard deviation (Agostinelli and Sorrenti, 2021). In contrast, less time with a parent can help or harm a child, because the effect depends on the care that replaces that time (Bernal and Keane, 2011; Bailey et al., 2021; Baker et al., 2019; Duncan et al., 2022). A year of informal child care in place of the mother's time lowered test scores by 0.14 of a standard deviation (Bernal and Keane, 2011), while Head Start raised college completion among the children who attended by 12 percentage points (Bailey et al., 2021). Head Start was an expansion that added to what a child received, and so were the cash and food programs that improved the adult health of the children they reached (Aizer et al., 2016; Hoynes et al., 2016). But the welfare reform of the 1990s was the opposite kind of change, a contraction that ended the cash entitlement and tied what remained to work. In the families that complied with the work rule, the reform raised earnings and replaced some of the parent's hours at home with other care. Much less is known about what a change of that kind does to children. The closest evidence comes from the 1993 expansion of the Earned Income Tax Credit (EITC), because that credit is paid only to families with earnings and so moved parents into work as the reform did. Agostinelli and Sorrenti (2021) find that children's short-term cognitive and behavioral scores fell under that expansion, and they trace the loss to the time mothers spent at work. Those scores were measured in childhood, and whether the loss lasts into adulthood is not known. This paper asks whether the young children of the 1990s reform were better or worse off as adults, once the earnings the rules added and the parent's time they took are counted together. It also asks whether the answer changed where a state's rules were stricter.

Rules that tie cash assistance to work are now the standard form of welfare in the United States. The reform of the 1990s is where they began. Between 1993 and 1998 every state replaced Aid to Families with Dependent Children (AFDC), an entitlement to cash for poor families with children, with aid that a parent had to work for and that ran out after a fixed number of months. States moved first under waivers from the federal program and then under Temporary Assistance for Needy Families (TANF), which Congress created in 1996. As the rules changed, welfare participation among single mothers fell by 23 percentage points between 1993 and 2002 and their employment rose by 11.3 percentage points (Fang and Keane, 2004). Studies that estimate the effect of the reform itself find the same move into

work among less-educated single mothers and a rise in their earnings and income (Kaushal and Kaestner, 2001; Bollinger et al., 2009). In my tabulations of the Current Population Survey (CPS), the employment of less-educated single mothers rose from 52 to 67 percent between 1993 and 1999, and their annual hours of paid work rose from 989 to 1,268 (Figure 1). They were the group most likely to receive welfare, while only 1 percent of higher-educated married mothers had welfare income. Over the same years the employment of these married mothers rose from 73 to 75 percent, and their annual hours of paid work from 1,306 to 1,355. A mother's time is a fixed endowment, so hours that move into market work are reallocated away from non-market time, including care of her child. On the children's side, nearly half of the children on welfare when the rules changed were under six (Herbst, 2017). For them the reform landed in the years when a family's cash matters most for later health (Hoynes et al., 2016) and when a parent's time matters most for a child's development (Del Boca et al., 2014).

Most studies of what the reform did to children measure its effects in childhood, and they find harm (Leonard and Mas, 2008; Herbst, 2017; Kalil et al., 2023; Dave et al., 2021; Paxson and Waldfogel, 2003). The harms range from higher infant mortality (Leonard and Mas, 2008) to more delinquency and substance use in adolescence (Dave et al., 2021). By contrast, few studies follow the reform's children into adulthood, and their findings are mixed (Hartley et al., 2022; Corman et al., 2022; Dore et al., 2025). Of these studies, Dore et al. (2025) is the closest to this paper, because it follows children exposed before age five in the Panel Study of Income Dynamics (PSID). Among those children it finds more psychological distress and chronic conditions in adulthood. However, neither the childhood studies nor the adult studies measure how strict a state's work rules were, taken together. Instead, each childhood study examines one rule at a time or the reform as a whole (Leonard and Mas, 2008; Herbst, 2017; Kalil et al., 2023; Dave et al., 2021; Paxson and Waldfogel, 2003). The adult studies date a child's exposure from the year the state's reform began (Corman et al., 2022; Dore et al., 2025), or they compare the years before and after reform (Hartley et al., 2022). Either way, their estimates measure exposure only by time under reform, so they treat a state that only let families keep more of their earnings the same as a state that took the whole grant from a family that did not meet a work requirement. It is therefore not known whether a child's adult outcomes depend on how strict those rules were, and not only on how long they applied.

That question has gone unanswered because of three features of the setting, and the design of this paper is built around each of them. The first is that every state reformed between 1993 and 1998, so no state kept the old welfare rules and the comparison cannot be between reforming and non-reforming states. But states reformed in different years, so children born

in the same year spent different parts of their early childhood under reform. The estimates therefore compare children born in the same year in different states, so that anything all states shared in a given birth year drops out. I also add fixed effects for each of the nine census divisions (the Census Bureau's groups of neighboring states) in each birth year, so a change that hit every state in a division in the same year does not enter the estimate. The second feature is that the content of reform differed from state to state and, within a state, from year to year. In 1993, for example, California changed only its earnings disregard, so that working families kept more of their grant, while Michigan required work and excused no parent of an infant. Studies that date a child's exposure from the year her state first reformed give the two states the same treatment (Corman et al., 2022; Dore et al., 2025), because a date records when the rules changed and not what they demanded. This paper therefore scores what the rules demanded. Congress wrote the Personal Responsibility and Work Opportunity Reconciliation Act (PRWORA) to move parents into work, so the rules this paper scores are those that govern work. Five rules govern work: the work requirement, the earnings disregard, the time limit, the sanction for not complying, and the age of the youngest child below which a parent is excused from work (the exemption age). Each rule's score runs year by year from 0 at the pre-reform program to 1 at the strictest setting any state used. The five scores are combined into one strictness index with fixed weights that sum to one. Each weight comes from the size of the rule's effect on parental employment in the experimental and quasi-experimental literature (see Appendix A; Miller et al., 2000; Smedslund et al., 2006; Grogger, 2003; Fang and Keane, 2004; Looney, 2005). Weights run from 0.42 on the work requirement down to 0.04 on the exemption age. For each child, I then add up the months from conception to her fifth birthday that she spent under reform in her birth state. Each month is weighted by the value of the index in the calendar year it falls in. Measured in years, the weighted total is the child's strictness-adjusted years of exposure. Every estimate quoted below is per strictness-adjusted year. One such year is one year with an index of 1 or two years with an index of 0.5. In the preferred specification I compare each child only with children of the same birth year whose states reformed in the same year, so the strictness of their states' rules is what separates them.

The third feature is that the rules reached a child only through her family's welfare status. I therefore multiply each child's exposure by the probability that her family was on welfare. To predict this probability, I fit a gradient-boosted trees model to families in the March CPS before reform and apply it to the characteristics each PSID child's family had before she turned five. In effect, I compare children with a high predicted probability of being on welfare against children with a low one and ask whether the gap in their adult outcomes was larger or smaller where the rules were stricter. All the children in the comparison are

PSID children born between 1975 and 1999 and followed into adulthood. From age 18 they contribute about 60,000 person-waves. I measure their adult outcomes with three indices, for general health, mental health and employment, and on each index a higher value is better.

I estimate four difference-in-differences specifications. The first difference is between children with a high and a low predicted probability of being on welfare, and the second is between states with stricter and less strict rules. In the first specification, exposure is the years a child spent under any reform, no matter how strict the rules were. In the second, I incorporate the strictness of the rules into exposure, so that exposure is measured in strictness-adjusted years. The preferred specification is the third, in which I keep exposure in strictness-adjusted years and also compare each child only with children born in the same year in states that reformed in the same year. The fourth is the same as the third, but with the five rule scores in place of the strictness index.

In the preferred specification, the rules left children with a high predicted probability of being on welfare healthier as adults. For a child whose family was at the 90th percentile of the predicted probability of being on welfare, relative to a child at the 10th, each strictness-adjusted year of exposure raises the general health index by 0.05 standard deviations and the mental health index by 0.07. Both health estimates are significant at the 10 percent level and are larger with strictness-adjusted exposure than with raw exposure, so the health estimates capture how strict the rules were and not only how long the rules applied. In contrast, the employment index shows no effect for these children. Of the three parts of the employment index, the share of adults employed does not move either, and the estimates for hours and earnings are too imprecise to say whether they move. However, in the fourth specification (Appendix B), a more generous earnings disregard raised the employment index, while a lower exemption age reduced it. These two rule effects offset each other when the five rules are combined in the strictness index, and this offset is part of why the employment index shows no effect.

The estimates are valid only if the welfare rules are the only reason that the gap in adult outcomes between children with a high and a low predicted probability of being on welfare is larger in states with stricter rules. As a first check, I estimate the effect of exposure on children the welfare rules could not reach, those with a predicted probability of being on welfare of zero. Any other change that could bias the estimates by reaching all children in states with stricter rules would show up in that effect, but the effect is indistinguishable from zero. However, this check cannot rule out a change that reached only families with a high predicted probability of being on welfare and was larger in states with stricter rules. The most likely change of this kind is a refundable state EITC adopted in 1993 or later, because it could vary with the same two differences as the welfare rules. So I re-estimate without

the children whose window from conception to age five could include a year with that credit. Without those children, both health estimates keep about the same size and stay significant.

These results are relevant today because work requirements are being extended to programs that did not have them and to parents who were previously excused. The 2025 reconciliation law attaches one to Medicaid (Congressional Research Service, 2025b) and widens the one in the Supplemental Nutrition Assistance Program (Congressional Research Service, 2025a). The results suggest that the welfare reform rules that govern work did not harm the children they reached and slightly improved their adult health. However, these results are the net of the income gained and the time lost. They also reflect the whole package of those rules and the labor market and policies of the 1990s. So the results apply only with caution to the new work requirements three decades later.

The paper makes two contributions to existing work. The first is to measure the net effect of the trade-off between the earnings a parent's work adds and the time it takes from a young child, using the welfare reform of the 1990s. On the income side of that trade-off, earlier work finds long-run gains from expansions that gave poor families food or health insurance (Hoynes et al., 2016; Bailey et al., 2024; Boudreaux et al., 2016; Goodman-Bacon, 2021b; Miller and Wherry, 2019). On the same side, Dustmann et al. (2024) find lower schooling and earnings in adulthood among the children of refugee families in Denmark after a cut of about 30 percent in their transfers. On the time side, earlier work finds that the care replacing a parent's time decides the effect on the child, with poor substitutes lowering children's outcomes and good ones more than making up for the lost time (Bernal and Keane, 2011; Baker et al., 2019; Bailey et al., 2021; Duncan et al., 2022). The reform studied here moved both sides of the trade-off at once, because its work rules raised earnings and took a parent's time. The second contribution is a measure of how strict each state's rules were, whereas the studies of this reform's children in adulthood measure only when the rules began (Corman et al., 2022; Dore et al., 2025; Hartley et al., 2022). The measure dates a state's reform from the first year a work rule was in force statewide. Beyond the date, the measure scores every rule that governs work from 0 at the pre-reform program to 1 at the strictest setting any state used, so two states that reformed in the same year need not receive the same treatment. This scoring matters because the earnings disregard and the exemption age moved the employment index in opposite directions when the rules enter the model separately, and a reform date alone cannot separate those two effects. The five scores form one state-year strictness index, and that index can be attached to any child-level data that record a birth state and a birth year.

2 Policy Background

From 1935 to 1996, Aid to Families with Dependent Children (AFDC) paid a monthly cash grant to low-income families with children, and each state set the size of that grant. In 1994 the maximum grant for a family of three with no other income was $366 a month in the median state, $120 in Mississippi and $923 in Alaska.[1] That year AFDC paid a grant to 5.0 million families in an average month,[2] and in 1996 one single mother in four was in a family on AFDC (Fang and Keane, 2004). In contrast, federal law set who qualified and what a parent had to do to keep the grant. A family qualified if it had a child under 18 whose parent was absent, incapacitated, deceased or unemployed, and if its income was below the state's limit and its assets at most $1,000 (U.S. House Committee on Ways and Means, 1996, pp. 384–394). From 1988 the federal Job Opportunities and Basic Skills (JOBS) program required adult recipients to take part in work, education or training (U.S. House Committee on Ways and Means, 1996, p. 411). The requirement exempted a parent whose youngest child was under 3, along with a few smaller groups,[3] and in 1994 the parents of a child under 3 alone headed 40.5 percent of all the families on AFDC (U.S. House Committee on Ways and Means, 1996, p. 411). A state had to enroll the parents who were not exempt only when it had the money to serve them, so a state that spent little on JOBS enrolled few (U.S. House Committee on Ways and Means, 1996, p. 411). In addition, the law set a minimum participation rate, and in 1994 a state had to have at least 15 percent of the adults who were not exempt in an activity (U.S. House Committee on Ways and Means, 1996, p. 412). The exempt groups together were about half of all the adults on AFDC, so that minimum came to about 7 percent of all the adults on AFDC (U.S. House Committee on Ways and Means, 1996, p. 422). A state that missed the minimum could lose part of its federal matching funds, but the law let the penalty be waived, and by 1996 the penalty had been waived every time (U.S. House Committee on Ways and Means, 1996, p. 412). In practice, states reported that in 1994, 20.4 percent of the families on AFDC had a member in a JOBS activity (Blank, 2002). An activity could be schooling or job search rather than work (U.S. House Committee on Ways and Means, 1996, p. 411). So the work requirement bound few of the parents on AFDC, both in law and in practice.

[1] University of Kentucky Center for Poverty Research National Welfare Data, in 1994 dollars; the median is over the fifty states and the District of Columbia.

[2] Author's tabulation of the University of Kentucky Center for Poverty Research National Welfare Data, summed over the fifty states and the District of Columbia.

[3] The other exempt groups were parents who were ill, incapacitated or of advanced age, those needed at home to care for an ill family member, those already working 30 hours a week, pregnant women from the second trimester, children under 16 or in school full time, and recipients living where the state ran no program (U.S. House Committee on Ways and Means, 1996, p. 411).

Those federal rules were the same in every state until states began to obtain waivers. Under Section 1115 of the Social Security Act, a state could apply to the Department of Health and Human Services for a waiver, a permission to set aside parts of the AFDC rules (Blank, 2002). The Clinton administration encouraged waivers that applied statewide (Blank, 2002). By August 1996, 27 states had such a waiver in place (Blank, 2002). Most of these waivers made the work requirement stricter (Blank, 2002). Instead, some let a working family keep more of its earnings before the grant was reduced (Bitler et al., 2005). In August 1996 the Personal Responsibility and Work Opportunity Reconciliation Act (PRWORA) replaced AFDC with Temporary Assistance for Needy Families (TANF). Twenty-four states started their TANF programs by the end of 1996, and 25 more states and the District of Columbia started theirs during 1997.[4] California started its program last, in January 1998. TANF ended the entitlement to aid, so a state could choose which families it supported (Blank, 2002). In addition, the federal government paid each state a fixed block grant in place of the matching funds it had paid under AFDC (Blank, 2002). The block grant did not fall when the number of families on aid fell, from 5.0 million in 1994 to 2.2 million in 2000.[5] With fewer families to pay, the states needed less of the block grant for cash. The states spent the savings on services instead. Cash payments took 75 percent of spending on family welfare under AFDC, and 40 percent under TANF in fiscal year 2001 (Burke, 2003). Child care was the largest of those services (Burke, 2003), so part of what the reform spent went to the care of children whose parents it had moved into work. Besides the block grant, TANF had two federal rules, and every other rule was the state's to set. No family could receive federally funded aid for more than 60 months in a lifetime, although a state could exempt up to a fifth of its caseload from this limit or set a shorter one (Blank, 2002). A further rule required a state to have half of its adult recipients in work activities for about 30 hours a week (Blank, 2002; Burke, 2003).[6] A state that fell short of that half lost part of its block grant (Burke, 2003). Schooling could not count toward the first 20 of the 30 hours (Blank, 2002).

The strictness index of the introduction is built from the work requirement, the exemption age, the sanction, the earnings disregard and the time limit. Each of these five rules had one setting under AFDC and a range the states could choose after reform. Table 1 lists each rule's AFDC setting, the change the paper counts as a move toward work, and the states'

[4]The waiver and TANF dates, and the rule settings below, are from the summaries of the Office of the Assistant Secretary for Planning and Evaluation in the Department of Health and Human Services and from the Urban Institute's Welfare Rules Database, described in Section 3.

[5]Average monthly caseload, University of Kentucky Center for Poverty Research National Welfare Data, summed over the fifty states and the District of Columbia.

[6]The statutory rate was 50 percent by 2002. A state's required rate was reduced by the fall in its caseload since 1995, so the effective rate was lower (Burke, 2003).

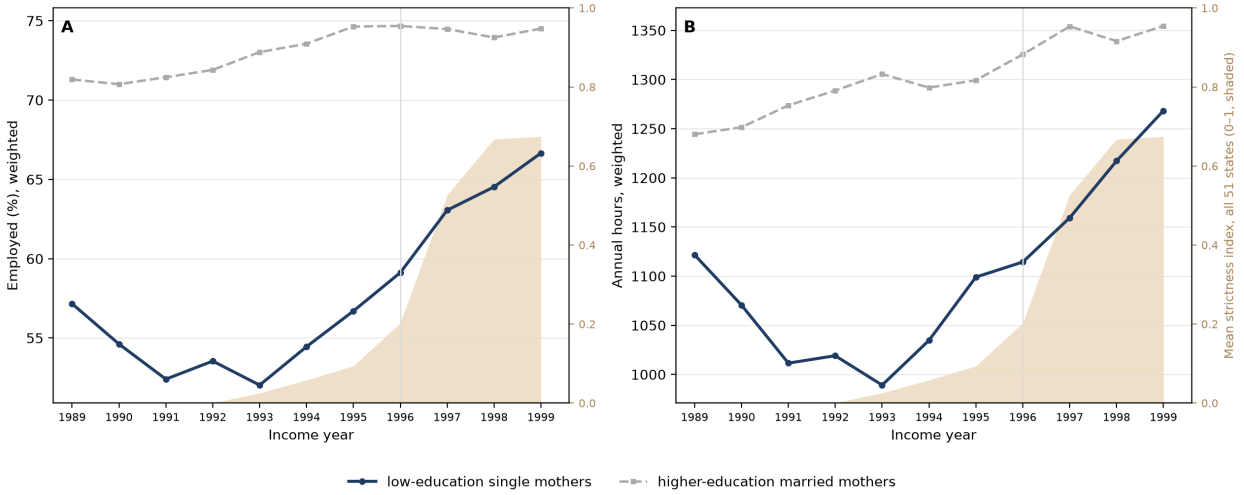


Figure 1: Employment and annual hours of mothers by likelihood of welfare receipt, March Current Population Survey, income years 1989 to 1999, using the survey's sample weights. Panel A is the share employed, measured as status in the survey week and plotted against the preceding income year. Panel B is annual hours, usual weekly hours times weeks worked in the income year, and zero for a mother who did not work. Mothers are women aged 18 to 54 with at least one child. The solid line is low-education single mothers, those with a high school diploma or less who are not married with a spouse present, 29.3 percent of whom had cash welfare income over income years 1989 to 1999. The dashed line is higher-education married mothers, 0.8 percent of whom did. The shaded area, on the right axis, is the mean strictness index across the 51 jurisdictions, and the vertical line marks 1996, the year PRWORA passed.

settings in 1998. The work requirement under AFDC was JOBS, and it bound few parents. Under TANF every state required recipients to work or to look for work as a condition of the grant (Blank, 2002). Under AFDC a parent was excused from the requirement until the youngest child turned three (U.S. House Committee on Ways and Means, 1996, p. 411). This exemption age was the state's to set under TANF, and a state could require work from the birth of the youngest child (Herbst, 2017). The sanction for a parent who did not comply removed the adult's share of the grant under AFDC and left the children's share in place (U.S. House Committee on Ways and Means, 1996, p. 524). A TANF state could remove the whole family's grant (Blank, 2002). The earnings disregard is the part of a family's earnings that did not reduce the grant. Under AFDC, in a family's first four months of work the disregard was $120 a month plus one third of the remaining earnings. For the next eight months it was $120 a month (Bitler et al., 2006). After those twelve months only $90 was disregarded, so each dollar earned above $90 cut the grant by a dollar (Bitler et al., 2006). Under a waiver or TANF a state could disregard far more, and from 1996 Connecticut disregarded all earnings up to the federal poverty line (Bitler et al., 2006). AFDC set no time

Table 1: The five rules under AFDC and the states' settings in 1998

Rule	Setting under AFDC	Move that counts as toward work	Settings across the 51 states in 1998
Work requirement	JOBS: participation in work, education or training; parent of a child under 3 exempt; 20.4 percent of families in a JOBS activity in 1994	A statewide requirement to work or look for work as a condition of the grant	All 51 required work or job search
Exemption age	36 months	A lower age	12 months in 27 states; below 12 in 19 (zero in five); 18 months in two, 24 in one, 36 in New Hampshire, 48 in Texas
Sanction	Adult's share of the grant removed; children's share kept	The whole family's grant removed	Whole family's grant in 35 states; adult's share in 16
Earnings disregard	\$120 plus one third of the rest for four months, \$120 for eight more, \$90 thereafter	A larger disregard (earnings kept before the grant is reduced)	Larger than AFDC's in 31 states, of which 11 disregarded earnings up to the poverty line; AFDC's rule in 20
Time limit	None	A limit shorter than 60 months	Shorter than 60 months in 20 states (18 to 48 months); the federal 60 months alone in 31

Notes: The AFDC settings are the federal rules of 1994 to 1996 (U.S. House Committee on Ways and Means, 1996; Bitler et al., 2006). The 1998 settings are the author's tabulation of the state-by-year policy panel described in Section 3, built from the summaries of the Office of the Assistant Secretary for Planning and Evaluation and the Urban Institute's Welfare Rules Database, over the fifty states and the District of Columbia. A move in the direction shown starts a state's reform year and raises its score on the strictness index; the exemption-age settings above 36 months (Texas) do not count as a move. Twenty-two states also had a family cap in 1998, which is not one of the five rules and does not enter the index.

limit, so a family could receive aid until its youngest child turned 18 (U.S. House Committee on Ways and Means, 1996, p. 385). Under TANF the federal limit was 60 months, and a state could set a shorter one (Blank, 2002). Where a state moved a rule away from its AFDC setting, the move pushed parents toward work, by making four of the rules stricter and the disregard more generous (Bitler et al., 2005).

I give each state one reform year, dated from when its waiver or its TANF program took effect. A state's reform year is the first year in which one of the five rules was in force

across the whole state at a setting different from the one AFDC had.[7] The setting must also have moved in the direction that pushed toward work. That direction is stricter for the work requirement, the exemption age, the sanction and the time limit, and more generous for the disregard, because a larger disregard raised what a working family kept and so made work pay (Bitler et al., 2005). A rule that took effect after June counts from the following year. For example, California's waiver raised the disregard in December 1992, so California is dated 1993. Because a state is dated only from the five rules, a waiver that only added a family cap does not start its reform year. A family cap denied a family an added grant for a child born while the family was on aid, so it changed the grant after a birth and not what the parent had to do to keep it. Georgia's waiver of 1994 was such a waiver, so Georgia is dated from its TANF program of January 1997. Nor does a waiver that ran in only a few counties start a state's reform year, because it did not reach all of the state's children. A state with such a waiver is dated from the year a rule applied statewide. The same rule dates every one of the states. Of the fifty states and the District of Columbia, three began in 1993, three in 1994 and three in 1995. Eight more states began in 1996, twenty-four states began in 1997, and the last ten began in 1998. Figure 2 plots these counts beside the years in which the waivers and the TANF programs took effect. In these counts, every state dated before 1997 was dated from a waiver, because no TANF program could start before PRWORA passed in August 1996. The ten states dated 1998 started their TANF programs after June 1997.

By 1998 every state had reformed, so the rules in force in that year show how far each state had moved from the AFDC rules. All 51 states required recipients to work or look for work. On the exemption age, 27 states set it at 12 months and 19 states set it lower. Five of those 19 states set it at zero, so a parent was expected to work from the month the child was born.[8] The remaining five states set it above 12 months. Two states set it at 18 months and one state at 24, New Hampshire kept the AFDC age of 36 months, and Texas raised it to 48. Neither New Hampshire's setting nor Texas's is a move toward work, so both states are dated from the first of their other rules that moved toward work. On the sanction, 35 states took the whole family's grant when a parent did not meet the work requirement, and 16 kept the AFDC rule of removing only the adult's share. On the disregard, 31 states disregarded more earnings than the $90 that AFDC allowed after twelve months of work, and in 11 of them a family of three kept part of its grant until its earnings reached the federal poverty line. The other 20 states kept the AFDC disregard rule. On the time limit, 20 states set a limit shorter than the federal 60 months, from Delaware's 48 down to Tennessee's 18, and

[7]The dates are the implementation dates in the ASPE summaries, not the approval dates.

[8]The 1998 counts in this paragraph are the author's tabulation of the state-by-year policy panel described in Section 3.

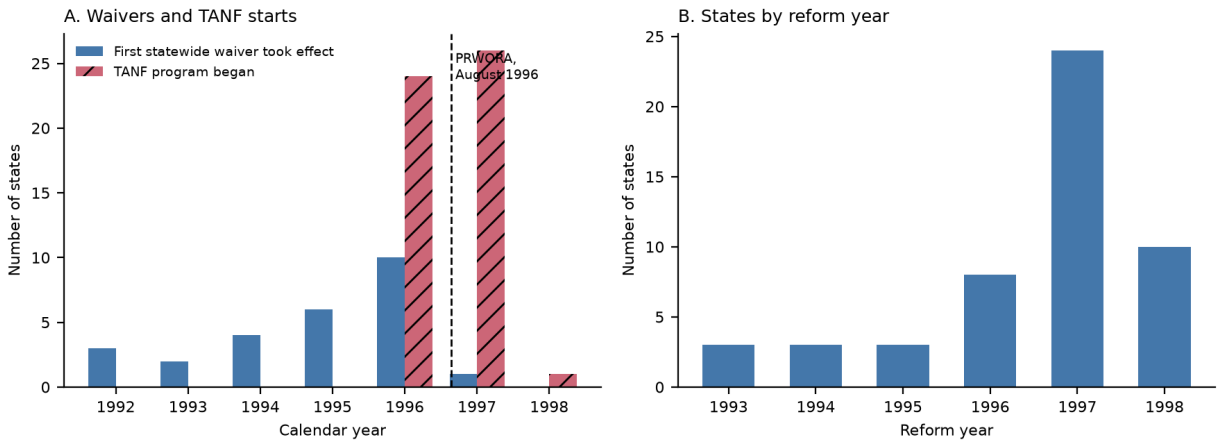


Figure 2: Waivers, TANF starts and reform years, 1992 to 1998

Notes: Panel A counts, by calendar year, the states whose first statewide waiver took effect and the states whose TANF program began; the dashed line marks August 1996, when PRWORA passed. Panel B counts states by the reform year this paper assigns, the first year in which one of the five rules was in force statewide at a setting that moved toward work, with a change after June dated to the following year. Both panels cover the fifty states and the District of Columbia. Dates are the implementation dates in the summaries of the Office of the Assistant Secretary for Planning and Evaluation.

the other 31 states applied the federal limit alone. So two states with the same reform year could differ in which rules had moved and in how far each rule had moved, because a state's reform year is the year in which its first rule moved toward work.

Apart from the cash grant, the reform left most of a family's other aid in place. PRWORA required states to keep Medicaid for any family that still met the AFDC eligibility rules of 1996, so losing the grant did not by itself end a family's eligibility for Medicaid (Bitler et al., 2005). Food stamp rules for families with children did not change (Bitler and Hoynes, 2016). So for a family on welfare the reform changed the rules that tied its grant to work, and little else. States changed those rules in different years, and how far each of the five rules had moved from the AFDC setting differed across states and from year to year. In Section 3 I score each rule from 0 at the AFDC setting to 1 at the setting that pushed hardest toward work, and combine the five scores into the strictness index. In Section 4 I compare children who were born in the same year and whose states reformed in the same year, so that what differs between them is the strictness index of their birth states in the years from conception to age five.

3 Data

Given the state-by-year timing of reform, the estimates require three things for each child: where and when the child was born, what the family looked like during the first five years

of life, and the child's health and employment as an adult. The Panel Study of Income Dynamics (PSID) records all three for the same person. It began in 1968 with about 5,000 families, 1,900 of them drawn from a low-income sample. Since then it has interviewed the members of those families and their descendants in 43 waves through 2023. The interviews were annual until 1997 and biennial since, so the later waves hold fewer adult observations per person. The spacing does not threaten the treatment, because the rules a child faced are dated from a state-by-year policy panel, built below, and not from the interview years. Two parts of the file supply the first of the three. State of residence is recorded in every wave. A childbirth history collected since 1985 adds the parent's report of each child's month and state of birth.

Two other surveys follow children from early life into adulthood, and neither carries birth state, early-childhood family and adult outcomes together. The children of the National Longitudinal Survey of Youth 1979 are born to one cohort of mothers, so a child's birth year and the mother's age move together and the two cannot be told apart. That survey also releases state of residence only under a restricted agreement. The only other candidate, the National Longitudinal Study of Adolescent to Adult Health, observes no child before adolescence and no state on its public file. What the PSID charges for carrying all three is the size of its usable sample. The restrictions set out below leave 8,408 children, and only 2,799 of those were exposed to reform.

3.1 Sample

Those restrictions start from every person in the PSID individual file born between 1975 and 1999, 25,942 people. The window opens in 1975 so that the oldest cohorts finished early childhood well before the first state reformed. It closes in 1999 because nothing is gained by going further, and something is lost. A child born in 1999 is the earliest whose whole exposure window, from conception to the fifth birthday, falls after reform in every state, so by that year the last states to reform, in 1998, already contribute a fully exposed cohort as the earliest states do. A child born later would not yet be 24 in the 2023 wave, the age from which the economic outcomes are measured. Within that birth window, four restrictions bring the 25,942 down to 8,408. The first keeps the 19,614 members of the two samples the PSID began with in 1968, the national cross-section and the low-income oversample, and drops the families the PSID added in 1997, 1999 and 2017 to represent immigrants, since PRWORA cut the eligibility of non-citizens rather than attaching conditions to their benefits, so those children faced a different treatment from the one this paper measures. The second leaves 19,582, dropping the 32 children born outside the 50 states and the District of

Columbia, because no state's rules applied to them at birth. The third keeps the 14,020 who appeared in an interviewed household before age 18, because the birth state and the family characteristics that predict welfare receipt come from that childhood record, and a person who joined the panel as an adult, most often by marrying into a PSID family, has neither. The fourth keeps the 8,408 who were observed at age 18 or older, so that an adult outcome exists, and 6,544 of them are observed at 24 or older, the age from which the economic outcomes are measured. Those restrictions also settle how many children the reform could have reached. For 2,799 of the 8,408, none of them born before 1988, some part of early childhood fell after the birth state's reform. For 905 of those, all of it did.

3.2 Birth state and month

Each child's birth state is the state whose rules define exposure. It comes from one of four sources, taken in order. For 5,956 children, 70.8 percent of the sample, the family was interviewed during the child's first year, and the state of residence at that interview is used. For 945 more the parent's childbirth history reports the state of birth. At the third rank are 1,504 children who were neither interviewed in the first year nor covered by the childbirth history. For them the state is that of the relative's household through which the child is linked to the panel. Three children have none of these and take the first state observed at age 2 or older. No child in the sample is without a birth state.

Two things put residence in the first year above the parent's report. It was recorded at the time rather than recalled. It also measures where the family lived, which is what the rules applied to rather than the place of delivery. Where the first two sources both exist they disagree for about one child in eight, because families move around the time of a birth and because some deliveries take place across a state line. The borrowed address at the third rank, checked against the parent's report where both exist, agrees about as often as an observed residence does, so it is kept.

Exposure is counted in months, so each child also needs a birth month. It comes from the childbirth history where it is reported and is set to June otherwise. Both the birth state and the birth month carry error. Disagreement between the sources assigns some children the rules of the wrong state. A birth month set to June puts some months on the wrong side of a reform date. Each of these is measurement error in the exposure variable. Such error moves the estimates toward zero rather than away from it, wherever it is unrelated to a child's true exposure.

3.3 The strictness index

The rules of the birth state are measured on a state-by-year index, one value for each of the 51 jurisdictions and each year from 1992 to 1998. It starts in 1992, the last year in which every state still ran the AFDC rules, so the index is 0 everywhere in that year. It ends in 1998, by which time every state had reformed. The index is built from the waiver and TANF summaries published by the Office of the Assistant Secretary for Planning and Evaluation and checked cell by cell against those documents. Children born in the late 1990s have windows that run past 1998, so the index is extended beyond that year from a second source. Each state's value is carried forward and changes only where the Urban Institute's Welfare Rules Databooks record a change. Those later changes are small, 17 state-years moving by 0.08 on average, against an average move of 0.51 in the year a state's reform began.

Five rules make up the index, each scaled so that 0 is the AFDC setting and 1 is the furthest any state moved away from it. Four of the five moved toward strictness and one, the disregard, moved toward generosity. The scale measures the distance in either case. Taking the five in turn, the work requirement is 1 when a statewide mandate was in force and 0 otherwise. The earnings disregard is scored by the earnings at which the benefit reaches zero. Under the AFDC rule that point was the statutory 90 dollars a month plus the monthly benefit, which scores 0. The scale rises to 1 where a family kept part of its grant up to the federal poverty line. The time limit scores 0 for AFDC, which set no limit at all, and for any state limit of 60 months or longer. Neither shortens the time a family may receive aid below the federal 60 months. It scores 1 at the 18 months Tennessee adopted, rising linearly as the limit falls between the two. The sanction is 1 when the whole family's grant was removed for non-compliance and 0 when only the adult's share was. The exemption is the age of the youngest child below which a parent is excused from work. Its score is 0 at the 36 months AFDC allowed and 1 where no age excused a parent, rising linearly as the age falls. Those five scales enter with weights that sum to one before rounding: 0.419 for the work requirement, 0.262 for the disregard, 0.157 for the time limit, 0.126 for the sanction and 0.037 for the exemption. Each starts from the consensus effect of that rule on parental employment in the experimental and quasi-experimental studies listed in Appendix A. Those five are not the only rules the states changed. The family cap is recorded but carries no weight, because its consensus effect on parental employment is close to zero and employment is the margin the index is anchored to.

Applied to every state and year, the five rules produce Figure 3. Before a state's reform its index is 0 in every year. In 1998 the index averaged 0.668 across the 51 jurisdictions with a standard deviation of 0.128, and ran from 0.439 in Alaska to 0.977 in Connecticut.

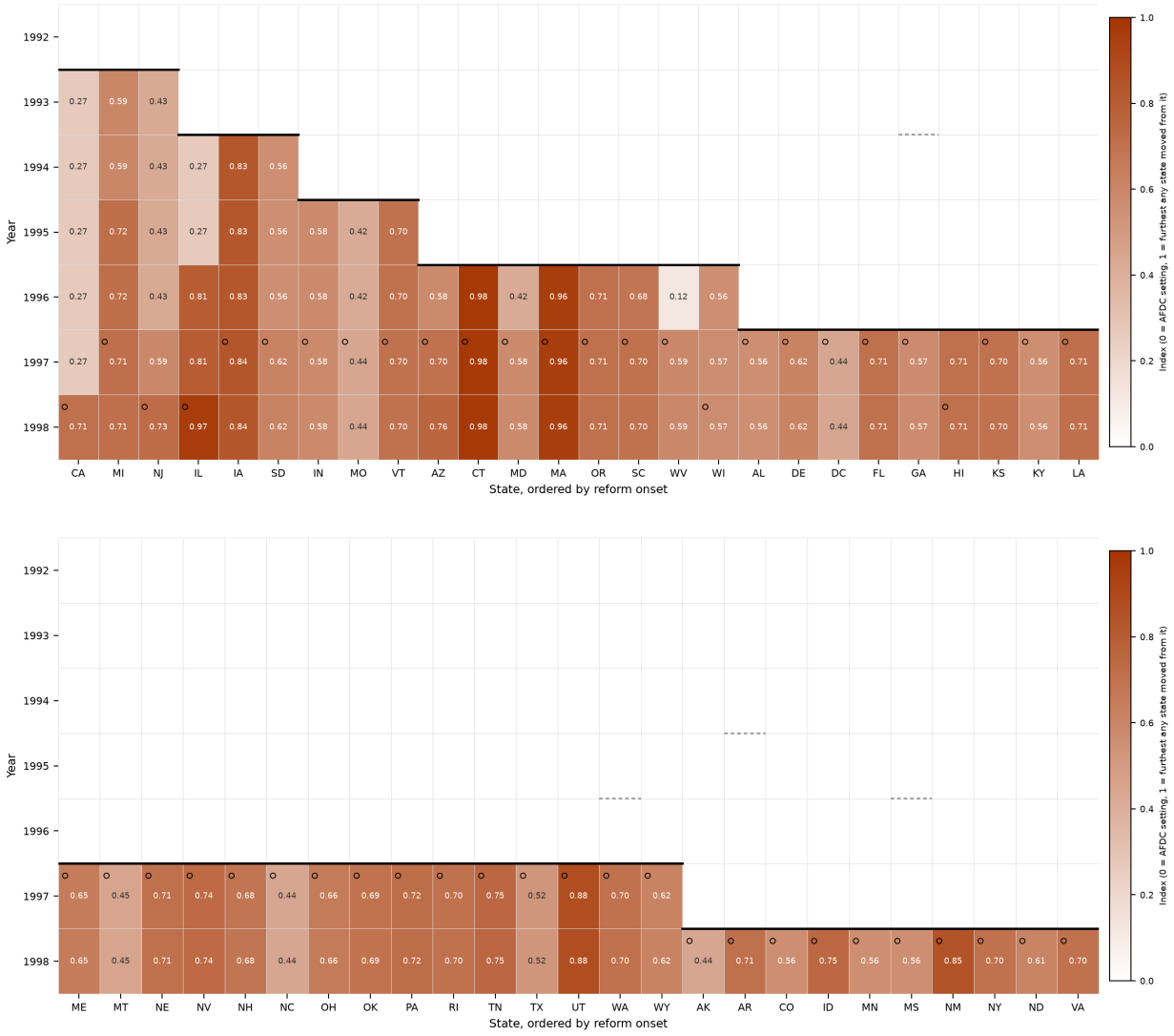


Figure 3: Strictness index by state and year, 1992 to 1998. The upper panel holds the 26 states that reformed earliest and the lower panel the remaining 25, each ordered by the year of reform. Rows are years and columns are states. The index combines five work rules with fixed weights, and is 0 at the AFDC setting and 1 at the furthest any state moved away from it; blank cells are 0. A black edge marks the first year in which one of the five rules had moved away from its AFDC setting across the whole state, a circle marks the first TANF year, and a grey dashed edge marks a state that left the AFDC rules earlier on a waiver that changed no work-related rule. Appendix D maps the year each state reformed and the mean index it ran from that year through 1998.

3.4 Exposure

The index measures how far a state moved. Dating when it moved is a separate step. A state's reform is dated from the first year in which one of those five rules had moved away

from its AFDC setting across the whole state, so a waiver that changed only the family cap, or that ran in a few counties, does not start the clock. The index takes one value per year, and a state's reform starts in January of its reform year. A rule that took effect in the first half of a year is dated to the whole of that year, and a rule that took effect after June is dated from the following January.

With the state's date fixed, the child's side of the measure is a window of 69 months, running from the month of conception, taken as nine months before birth, to the month of the fifth birthday. Exposure is the share of those months that fall on or after the birth state's reform date. Strictness-adjusted exposure replaces each reformed month by the index value in force in the birth state that year and divides the sum by the same 69 months, so it equals exposure when every reformed month carried an index of 1 and is lower whenever the state's rules sat below the maximum. Dividing that same sum by 12 instead of 69 gives the child's strictness-adjusted years of exposure, the unit in which this paper quotes every estimate.

Across the 8,408 children the mean exposure is 0.220. Among the 2,799 exposed children the mean strictness-adjusted exposure is 0.411, with a tenth percentile of 0.082 and a ninetieth of 0.714. In years that mean is 2.4, and the 905 children exposed for the whole window average 3.8. These are the means for one set of cohorts, and they say nothing about how exposure built up over calendar time. Figure 4 shows that build-up in a different survey, for all children aged 0 to 5 in the Current Population Survey. By 1999 the average child there had spent 87 percent of life so far under reform. Weighting each month by the index in force brings that share to 56 percent.

3.5 Predicted welfare receipt

The rules reached a child only through a family that was on welfare or would have been, so the design needs each child's chance of being in such a family, estimated from behaviour before the rules changed. Whether the family actually claimed AFDC cannot serve that purpose. A claim is itself a response to the rules, so for a child born in 1996 the family's receipt is measured after her state reformed and is an outcome of the treatment. Taking receipt from the years before any state reformed, 1990 to 1992, does not solve this, because those years fall before the birth of the youngest cohorts and a decade after the early childhood of the oldest, so the measure would stand for a different stage of life in every cohort, and which stage it stood for would itself be determined by the child's birth year and so by her exposure. And two families with the same characteristics still differ in whether they claim, a willingness that may reach the child through paths other than the rules.

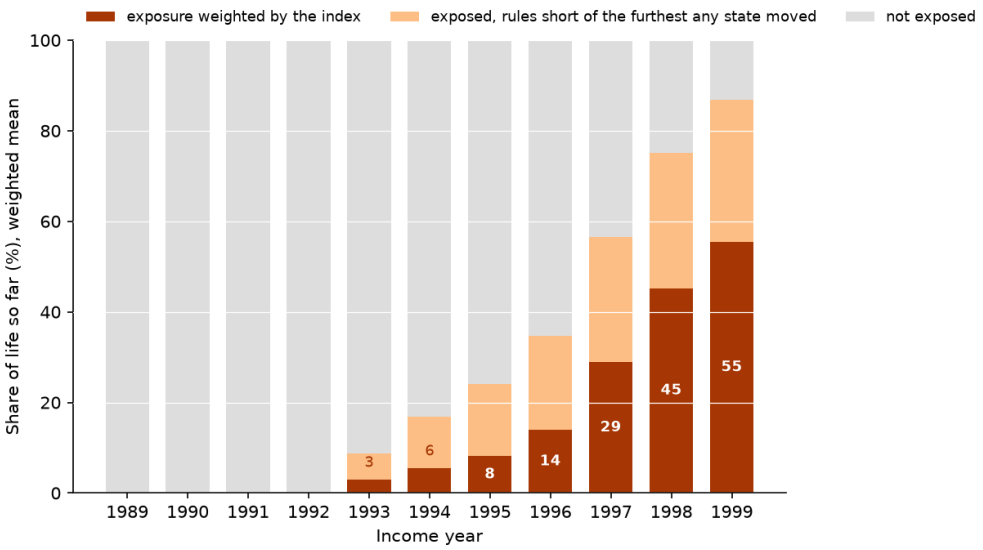


Figure 4: Share of life so far lived under reform, children aged 0 to 5 in the March Current Population Survey, by income year, using the survey's sample weights. The dark bar weights each year of life under reform by the index in force in that year, the light bar is the remainder of the years under reform, and the grey bar is the share of life not under reform. The figure prints the dark bar's value. This is a repeated cross-section of all young children, not the PSID sample used for the estimates.

Hoynes et al. (2016) face the same difficulty of an endogenous claim and answer it with a participation rate for each of twelve family types, measured before the program arrived. Each child here carries a predicted probability, P, that a family with her family's characteristics received AFDC before any state reformed, estimated for the individual family rather than for a group of them. The unit predicted for is the family rather than the household, because AFDC paid a parent and her children and not a dwelling. Eight characteristics enter: whether the head finished 12 years of school, the head's race, whether the head was married with the spouse present, the head's age, the size of the family, the number of own children under 18, the number of other adults in the unit, and the state. Income, employment and hours are left out because they are the reform's own channel, the very thing the rules were meant to move. Receipt of other programs is left out because a claim in one program carries the same willingness to claim that the prediction is meant to exclude. The characteristics are read over the child's own ages 0 to 5, the years of early childhood the exposure window covers apart from the nine months before birth.

The model that maps those characteristics onto receipt is a set of gradient-boosted trees. It returns a prediction when a characteristic is missing, so no child loses a value. Fitting such a model needs families observed before any state reformed, with their characteristics and their welfare receipt both recorded, and enough of them for the relationship between the two to

be estimated. The March Current Population Survey supplies them. It is a national survey that records the previous calendar year's income, program receipt and family composition, and its 1984 to 1993 surveys cover income years 1983 to 1992, all before the first state reformed. Those surveys hold 102,884 families with a child under 6 headed by the mother or her partner. The model is fitted to them. Fitting it there also leaves the PSID free to serve as an independent test. It is then applied to every child in the sample, exposed or not.

One group is the exception, and needs a model of its own. AFDC counted a grandparent's income against the grant of a mother who lived in her parents' household, so these mothers received aid less often than their own characteristics alone would imply. The main model predicts about twice the receipt they actually report. They are scored instead by a logit on the mother's race, her age and her number of own children, fitted on the 1,507 families of this kind in the same surveys. Those 1,507 are a separate set of families from the 102,884, filed by the survey as second family units within a household, and they take no part in fitting the main model.

The prediction is then tested against the PSID's own record. PSID families interviewed between 1990 and 1993 with a child under 6, all before reform, are scored by the model, and each score is set against whether the family reported receiving AFDC. The model was fitted without any PSID record, so this is an out-of-sample test. The area under the receiver operating characteristic curve (AUC), the probability that a recipient family is scored above a non-recipient family, is 0.903. On the scale of Hosmer et al. (2013), an area of 0.7 to 0.8 is acceptable discrimination, 0.8 to 0.9 excellent, and 0.9 or above outstanding, so the model sits in the top band. Those same scores spread widely. Weighted by the PSID longitudinal weight, the tenth percentile is 0.008 and the ninetieth is 0.460. That spread is what lets P sort children within a state and birth year by how likely the rules were to reach them, and Section 4 states the assumption under which that sorting identifies the effect.

3.6 Outcomes

The PSID asks its health and economic questions of the head of the household and the spouse, so a grown child living in a parent's home has no outcome of her own in the main files. The Transition into Adulthood Supplement fills those years. It has been fielded since 2005 to members aged 18 to 28, whether or not they head their own household, so it reaches the people the main interview misses. A supplement value is used only where the main file has none for that person in that year. The supplement adds 868 people who would otherwise carry no adult health report at all. With it 6,439 people carry at least one self-rated health report, against 5,571 without it. Those 868 are the youngest people in the sample, with a

median birth year of 1995 against 1986 for the rest, so they are also among the most exposed to the reform. Without it, the cohorts the design most needs would be the worst covered.

What those people are measured on follows from what the reform changed in a family, its cash and the parent's time at home, during the years in which adult health is thought to be set, and from what the reform set out to do. Self-rated health and obesity are among the outcomes that Hoynes et al. (2016) and Boudreaux et al. (2016) found to move with early-childhood income and insurance. Those papers work through the family's resources. The reform also moved the parent's time, and conditions on aid could reach a young child through the hours a parent spent at home and through the strain of meeting them. The distress scale and the psychiatric diagnosis are the outcomes that would carry a mark of that second channel in the grown child. The employment outcomes are included because work was the reform's stated aim, so a change in the child's own adult work would be the reform reaching the next generation through the same margin.

Each of those outcomes enters oriented so that a higher value is better. For the two bad events below, that means the analysis variable is the absence of the event. The health outcomes are measured from age 18. Self-rated health is the survey's five-point scale, reversed here so that 5 is excellent, and available for these cohorts from 1990. Obesity is a body-mass index of 30 or more from reported height and weight, asked from 1999. It enters as the share not obese. Whether a doctor has ever diagnosed a psychiatric problem is asked from 1999 as well, and enters as the absence of a diagnosis. Psychological distress is the six-item Kessler scale, asked from 2001, scored as the share of its 24 points the person does not report. The economic outcomes are measured from age 24. They are employment, annual hours and own labor earnings, and earnings enter in the inverse hyperbolic sine so that zeros stay in. Those single outcomes are also combined. Section 5 reports three summary indices alongside them: a general health index from self-rated health and obesity, a mental health index from the distress scale and the psychiatric diagnosis, and an employment index that combines employment, hours and earnings. Each is the average of its components' z-scores, standardized on the children with no exposure. An inverse-covariance-weighted version of each index (Anderson, 2008) is correlated at 0.985 or more with the equal-weight one, so the two weightings rank children almost identically.

3.7 Summary statistics

With those variables defined, Table 1 compares the exposed and the unexposed children. The exposed were born in 1995 on average and the unexposed in 1983, since exposure begins with the 1988 cohort. The two groups differ on every characteristic that enters P. A parent

Table 2: Summary statistics by exposure to reform

	Unexposed	Exposed	All
Panel A. Characteristics (one row per child)			
Birth year	1982.6	1994.7	1986.7
Predicted welfare receipt P	0.167	0.228	0.187
Parent with less than 12 years of school	0.237	0.265	0.247
Nonwhite	0.454	0.514	0.474
Male	0.502	0.517	0.507
Share of ages 0 to 5 with mother married	0.693	0.539	0.642
Family income to needs, early childhood	2.66	2.86	2.73
Exposure share	0.000	0.661	0.220
Strictness-adjusted exposure	0.000	0.411	0.137
Strictness-adjusted years of exposure	0.00	2.36	0.79
Children	5,609	2,799	8,408
Panel B. Health outcomes, age 18 or older			
Self-rated health (5 = excellent)	3.69	3.65	3.68
Not obese	0.694	0.722	0.700
Distress scale (1 = no distress)	0.824	0.783	0.817
No psychiatric diagnosis	0.877	0.849	0.873
Person-waves	47,118	12,612	59,730
Panel C. Employment outcomes, age 24 or older			
Employed	0.780	0.753	0.776
Annual hours	1,726	1,695	1,722
Own labor earnings, inverse hyperbolic sine	9.60	9.31	9.56
Person-waves	31,059	5,256	36,315

Notes: Means, unweighted. The sample is the 8,408 PSID children born 1975 to 1999 described in Section 3. A child is exposed when any month of the 69-month window from conception to the fifth birthday falls on or after the birth state's reform date. Exposure share is the fraction of those months falling on or after the reform date; strictness-adjusted exposure weights each such month by the strictness index in force, which is 0 at the AFDC setting and 1 at the furthest any state moved from it. Strictness-adjusted years divide the same sum by 12 rather than 69. Predicted welfare receipt P is the modelled probability that a family with the child's early-childhood characteristics received AFDC before any state reformed. Panel A has one row per child; family characteristics are measured over the child's ages 0 to 5, and Nonwhite and Male refer to the child. Panels B and C report person-waves rather than children, pooling every survey wave in which the person is observed at the stated age, from the core interview or the Transition into Adulthood Supplement, so a person contributes as many rows as the waves in which she is observed. Every outcome is oriented so that a higher value is better. The distress scale is one minus the six-item Kessler score divided by 24. Earnings enter in the inverse hyperbolic sine, the form used in Section 5.

with less than 12 years of school is found for 27 percent of the exposed and 24 percent of the unexposed, and 51 percent of the exposed are nonwhite against 45 percent. Predicted receipt averages 0.228 among the exposed against 0.167. The largest of these gaps is the share of ages 0 to 5 with the mother married, 54 percent against 69 percent. Table 1 then turns from the children to their adult outcomes, which are measured once per survey wave rather than once per child. On the table's orientation, in which a higher value is always better, the exposed report 3.65 against 3.69 on the five-point self-rated health scale. Their person-waves are more often free of obesity, 72 percent against 69. On the distress scale, where a higher value means less distress, they sit at 0.78 against 0.82. Fewer of their person-waves are free of a psychiatric diagnosis, 85 percent against 88. At 24 or older they are employed slightly less often, 75 percent against 78 percent, and work about the same hours, 1,695 a year against 1,726. The exposed are twelve years younger than the unexposed and were interviewed in later waves, so every one of these comparisons mixes the reform with age and with calendar time. The estimates in Section 5 therefore never compare an exposed cohort with an unexposed one directly, but compare children of the same birth year across states and, within a state and birth year, across levels of P.

4 Empirical Strategy

4.1 The estimate, built in four steps

I build the estimate in four steps, and each step adds one thing to the specification before it. The first measures reform only by the months a child spent under it. It compares the adult outcomes of children who spent more of their early childhood under reform with those who spent less, and asks whether that comparison looks different for children whose families were more likely to be on welfare. This makes each step a difference-in-differences, with the first difference between children with a high and a low predicted probability of being on welfare and the second between children with more and less exposure to the rules. For child i born in state s in year b and observed in survey wave t,

$$y_{isbt} = \beta_1 E_i + \beta_2 E_i P_i + \gamma P_i + \alpha_s + \lambda_b + \phi_{d(s)b} + \psi_t + X_{it}'\delta + \varepsilon_{isbt}. \tag{1}$$

Here y is an outcome from the list in Section 3, oriented so that a higher value is better. E_i is the share of the 69 months from conception to the fifth birthday, nine in the womb and sixty after birth, that fell under any reform in the child's birth state, so it runs from 0 for a child whose window closed before her state reformed to 1 for a child whose window opened after it. P_i is the predicted probability that a family like the child's received AFDC

before any state reformed. β_1 is the effect of moving E_i from 0 to 1 on a child whose family had no predicted chance of receipt, and β_2 is how much larger or smaller that effect is for a child whose family was certain to be on welfare. β_2 is the coefficient of interest, and it is an intention-to-treat effect, since P is a probability and not the family's actual receipt. A family with a high P that never took up aid counts as exposed all the same, so the estimate is smaller in size than the effect on the families the rules actually reached.

Each fixed effect absorbs one kind of difference that would otherwise be counted as an effect of exposure. The birth-state effects α_s absorb everything fixed about a state, including its benefit level before reform and the health of its population. The birth-year effects λ_b absorb everything a cohort shared across the country, including the national part of the 1990s boom and the federal EITC expansions. The census-division by birth-year effects $\phi_{d(s)b}$, with $d(s)$ the census division of state s, absorb whatever a cohort shared with the other states of its division, so a division whose children were getting healthier across cohorts faster than another's contributes nothing to β_2. They also fix who is compared with whom, because a child is now compared only with the other children of her division born in the same year. The survey-wave effects ψ_t absorb the year in which an outcome was measured. Two terms are left. The main effect of P is γ, and it absorbs the level difference in adult outcomes between children of likely-recipient families and the rest. The controls X_{it} are the child's sex, race, age and age squared, and the log of the family's income-to-needs ratio over the same 69 months. δ is the vector of coefficients on them, and the last term, ε_{isbt}, is the error term.

Within a division and a birth year, β_2 can use every way in which exposure and the chance of receipt differ between children. Exposure differs between the states of a division because they reformed in different years, and within a state it differs across cohorts because a later cohort spent more of its window under the rules. The chance of receipt differs in a third way, between families within a state and cohort. β_2 draws on all three at once, through how the gap in adult outcomes between likely-recipient children and the rest moves with exposure across the states and cohorts of a division.

What this first specification cannot do is tell one reform from another, because E_i counts a month the same whether the state required a parent to work or only raised the earnings disregard. Corman et al. (2022) and Dore et al. (2025) measure exposure this way, from the date a state's reform began, and their estimates are therefore of reform on average, across whatever the states chose to do.

The second step puts what the rules required into the exposure measure and changes nothing else. Each month of the window now takes the value of the strictness index in force in the birth state for the calendar year the month falls in, and a month before the state

reformed takes zero. W_i is the average of those 69 values, so it runs from 0 for a child never under reform to 1 for a child under the strictest rules for the whole window, and a year under rules scored 0.5 counts as half a year. W_i replaces E_i in equation (1), and β_2 then measures what the rules required and not only how long they applied. A state is still compared with every other state in its division, whatever year each reformed.

That comparison has a cost, because a state that reformed in 1993 is then a comparison for a neighbour that reformed in 1998 while its own effect on its children is still building. Goodman-Bacon (2021a) shows that an estimate pooled over such comparisons can weight them wrongly when effects build over time. The estimators that Callaway and Sant'Anna (2021) and de Chaisemartin and D'Haultfœuille (2020) built to avoid this compare each treated unit only with units not yet treated at the same date, and this sample cannot supply them. Every state reformed between 1993 and 1998, so no state is never treated, and a clean comparison can only be a state whose reform lies beyond a cohort's window. For the cohorts born from 1988 to 1992 the late reformers are such states. From the 1993 cohort on there are none, because the last state reformed in 1998 and the window reaches age five, so every state has reform months in the window of every later cohort. The only states whose exposure then holds still are the early reformers, at full exposure from the 1995 cohort, and a state already fully treated is the comparison those estimators are built to refuse. That leaves no clean comparison for any cohort born from 1993 to 1999, and those cohorts hold 77 percent of the exposed children. The third step therefore removes the comparison rather than correcting it. It replaces the birth-year effects λ_b with reform-wave by birth-year effects $\lambda_{w(s)b}$, where $w(s)$ is the year in which state s reformed. These narrow the comparison a second time. From here on, a child is compared only with children born in the same year in states that reformed in the same year as hers, on P across their families and on W across those states. For the same child i, the third step is

$$y_{isbt} = \beta_1 W_i + \beta_2 W_i P_i + \gamma P_i + \alpha_s + \lambda_{w(s)b} + \phi_{d(s)b} + \psi_t + X'_{it}\delta + \varepsilon_{isbt}. \tag{2}$$

Within a wave, states share the year of reform, so an early reformer is never a comparison for a late one. Goodman-Bacon (2021b) does the same in his study of Medicaid, with effects for the year a state adopted the program interacted with birth cohort, so that early and late adopters are never compared. I take this third step as the preferred specification, because the year a state reformed can no longer enter the comparison, and every number the paper quotes comes from it.

What remains to identify β_2 is the difference in P across families and the difference in strictness across states that reformed together. Inside a wave, the states reformed in the

same year, so for any cohort their children spent the same months under reform, and W differs between them only because their rules differed. A stricter state has a higher W for the same months. β_1 therefore compares how outcomes moved across cohorts in the stricter states of a wave with how they moved in the less strict ones, and any shock that moved with the strictness of the rules without being the rules goes into it. β_2 uses that same difference in strictness, but only through the gap in adult outcomes between children of likely-recipient families and the rest. A shock that moved all the children of a state and cohort alike changes the level of their outcomes and not the gap, so it enters β_1 and not β_2. The only shock that can reach β_2 is one that moved likely-recipient families relative to the rest, in step with the strictness of the state's rules. The identifying assumption is that no shock of that kind occurred, so among states that reformed in the same year, the gap in adult outcomes between children of likely-recipient families and other children born in the same year would have moved across states in the same way had the states' rules not differed. It would fail if something else that varied across states with the strictness of reform reached likely-recipient families alone, such as a state EITC adopted in the same years.

The reform years also fix which cohorts carry exposure. It begins with the 1988 cohort, the first that can carry any, because the first state reformed in 1993 and the window reaches age five. The sample reaches back to the 1975 cohort, so the thirteen cohorts born before 1988 carry no exposure and do not identify β_2, and I keep them because the fixed effects and the controls are more precisely estimated with them than without.

The fourth step keeps everything in equation (2) and replaces W_i with the five rules that make up the index, entered together, each with its own main effect and its own interaction with P_i. With W_i^k the share of the window that rule k covered, on the same 0 to 1 scale as W_i, the fourth step is

$$y_{isbt} = \sum_{k=1}^{5} \left(\beta_1^k W_i^k + \beta_2^k W_i^k P_i\right) + \gamma P_i + \alpha_s + \lambda_{w(s)b} + \phi_{d(s)b} + \psi_t + X_{it}'\delta + \varepsilon_{isbt}. \qquad (3)$$

β_2^k is the effect of rule k at its strictest setting over the whole window, with the other four rules held fixed. Nothing about the comparison changes at this step, and Appendix B reports it to show which of the five rules the result comes from.

4.2 What the tables report

Tables 2 to 4 report the first three steps as a column each, and each column gives β_2 in two rows. The first row is the coefficient itself, for a move in exposure from 0 to 1, the whole 69-month window as that column measures it. The second row rescales it to one year of

exposure and to the gap in P between a child at the 90th percentile and one at the 10th, and it is the row every per-year figure in the text comes from. I report the 90th against the 10th percentile rather than $P = 1$ against $P = 0$ because the endpoints are almost empty. The 90th percentile of P is 0.46, so a family certain to be on welfare sits above nearly every family in the sample, and a figure at that contrast would describe a comparison the data barely contain. The two percentiles are values of P that children in the sample actually have, so the figure describes a difference the data can show. The gap between them is 0.453, computed over children rather than person-waves, each child counted once and weighted by the PSID weight of Section 4.3, and the window is 5.75 years long, so the second row is the first row multiplied by 0.453, to go from the 0-to-1 contrast in P to the 90th–10th gap, and divided by 5.75, to go from the whole window to one year.

4.3 Inference

Standard errors are clustered on the birth state, because the treatment varies by state and cohort and errors can be correlated within a state across its cohorts and survey waves. That leaves only 51 clusters, the 50 states and the District of Columbia, so cluster-robust p-values can overstate precision, and for β_2 in the first three steps I therefore report p-values from a restricted wild-cluster bootstrap with 999 Rademacher draws, in which the null is imposed (Cameron et al., 2008). The fourth step carries cluster-robust p-values alone, because the bootstrap is run one coefficient at a time and that step has five interactions. Every regression also carries the PSID's individual longitudinal weight. The weight corrects both the survey's oversample of low-income households and the families it loses as the panel runs on.

4.4 Validity of the design

The threat the identifying assumption leaves standing is a shock to a state and cohort that reached likely-recipient families alone and coincided with the state's reform. The census-division by birth-year effects take out whatever part of such a shock was common to a division's states. What survives is a shock to one state's likely-recipient families that moved with that state's reform and not with its division's. The state EITC named in Section 4.1 is a shock of this kind. I test for it in Section 6 by dropping the children whose window from conception to age five could include a year with a refundable state EITC that started in 1993 or later. If the estimates hold without those children, a state EITC did not produce them.

The same regression gives a second check in its main effect of exposure. β_1 is that effect at $P_i = 0$, where a family had no predicted chance of receipt. The rules paid such families

nothing, and this check assumes that reform reached their children in no other way, such as through the local labour market. So exposure should not move the outcomes of those children, and if it does, something other than the rules moved with a state's reform across its cohorts. Hoynes et al. (2016) read a null of this kind, estimated on families unlikely to use food stamps, as support for their design, and I read β_1 the same way. What the check cannot detect is a shock that reached likely-recipient families alone, because such a shock moves β_2 and leaves β_1 at zero. A null on β_1 therefore leaves one shape of confound standing, a shock that moved with strictness and reached likely-recipient families alone, and the state EITC test above is the check on the one such shock the setting names. Within that limit, Section 6 reports β_1 in Table 5, for the ten outcomes of Tables 2 to 4, with and without the census-division by birth-year effects. The census-division effects have removed something that moved with a state's reform if β_1 is away from zero without them and at zero with them. If β_1 is away from zero with them, then something moved with one state's reform that the division effects do not reach. A shock of that kind enters β_1 rather than β_2, but it shows that the fixed effects did not remove everything that moved with that state's reform, and the assumption that nothing reached likely-recipient families alone becomes harder to hold.

5 Results

Tables 2 to 4 report the estimates for the three indices in the order Section 3 listed them, general health, mental health and employment, each with its components. Each table has the same structure, with one block per outcome and the first three specifications of Section 4 as its three columns. Within a block the first row is the interaction of exposure with P, β_2, with its clustered standard error and the wild-cluster bootstrap p-value, and the second is β_1, the coefficient on exposure alone and so the effect at $P = 0$. The third rescales β_2 to one year of exposure as that column measures it, a plain year in column 1 and a strictness-adjusted year in columns 2 and 3, for a child at the 90th percentile of P against one at the 10th, 0.453 apart on the P scale. I take column 3 as the preferred estimate because it is the column that addresses staggered adoption, by comparing children only with children of the same birth year in states that reformed in the same year, so the estimate comes from how strict those states' rules were and from the family's predicted receipt. Every number quoted in the abstract and introduction comes from that column. The one other convention in what follows is that a 90 percent confidence interval is built from the clustered standard error, so it can be narrower than the bootstrap p-value implies. The fourth specification enters the five rules in place of the index, and it is reported in Appendix Table B1. I return to it at

the end of this section.

Table 2 presents the estimates for the general health index and its two components, self-rated health and obesity. Under the preferred specification in column 3, each strictness-adjusted year of exposure raises the general health index of a child at the 90th percentile of predicted welfare receipt by 0.047 standard deviations relative to a child at the 10th percentile, a coefficient of 0.600 with a wild-cluster bootstrap p-value of 0.081. Of the two components, only self-rated health is significant, with a coefficient of 0.636 and a bootstrap p-value of 0.077. Per year at the same contrast that is 0.050 points on the five-point scale, 1.4 percent of the mean of 3.70. Not being obese moves the same way and is imprecise, rising 1.0 percentage point per year with a 90 percent confidence interval from -0.5 to 2.4 points. A child exposed for the whole window at the average strictness of the fully exposed, 3.8 strictness-adjusted years, therefore gains 0.18 standard deviations on the index at the same contrast in P. The closest published number to set that against is from Hoynes et al. (2016). In their full sample they estimate the effect of food stamps the way equation (1) does, with exposure interacted with the family's probability of using the program, and for a family certain to use it they find that full access from conception to age five lowered a metabolic syndrome index by 0.4 standard deviations. Their probability is one participation rate for each of twelve family types, whereas in this paper P is predicted for each family on its own, from eight of its characteristics, as Section 3 describes. Both put the effect at a family certain to use the program, so the sizes can be set side by side. Their full exposure means the program was in place for the whole window, and the children here with reform in place for the whole window faced rules worth 3.8 strictness-adjusted years on average. The matching number here is β_2 times 3.8 divided by 5.75 for a child whose family was certain to be on welfare, or 0.40 standard deviations on general health. The two indices measure different things, so the comparison says only that the effects are of a similar size.

Columns 1 and 2 of Table 2 show the two steps before the preferred estimate. With the months under any reform unweighted, the index coefficient is 0.349, and scoring the same months by strictness raises it to 0.520, so most of the rise across the three columns comes from measuring what the rules required. The reform-wave by birth-year effects of the third step then raise it by a further 0.080, to 0.600. Self-rated health follows the same path, 0.360, 0.578 and 0.636. Not obese stays between 0.077 and 0.122 and is not significant in any column. The bootstrap p-values for the index run 0.11, 0.12 and 0.08. The second row of each block holds the main effect of exposure, and in every column of Tables 2 to 4 it is indistinguishable from zero. The largest t-statistic is 1.6, on the employment index in column 1. Section 6 takes that row up as the design's placebo.

Table 3 turns to the mental health index, the distress scale and the share with no psychi-

Table 3: Exposure to welfare reform and adult general health

	(1) Share under any reform	(2) Strictness-adjusted share	(3) Column 2 with reform-wave × birth-year effects
General health index			
Exposure $\times P$	0.349 (0.188)	0.520 (0.289)	0.600* (0.276)
	[0.11]	[0.12]	[0.08]
Exposure	-0.001 (0.143)	0.058 (0.222)	-0.330 (0.404)
Per year of exposure, P 90th vs 10th pct.	0.027	0.041	0.047
Mean of outcome	0.057	0.057	0.057
Person-waves	40,731	40,731	40,731
Persons	6,225	6,225	6,225
Self-rated health (5 = excellent)			
Exposure $\times P$	0.360* (0.179)	0.578* (0.267)	0.636* (0.252)
	[0.10]	[0.08]	[0.08]
Exposure	0.022 (0.149)	0.179 (0.229)	-0.116 (0.360)
Per year of exposure, P 90th vs 10th pct.	0.028	0.046	0.050
Mean of outcome	3.704	3.704	3.704
Person-waves	40,713	40,713	40,713
Persons	6,224	6,224	6,224
Not obese			
Exposure $\times P$	0.077 (0.074)	0.090 (0.110)	0.122 (0.112)
	[0.34]	[0.46]	[0.32]
Exposure	-0.009 (0.047)	-0.039 (0.076)	-0.192 (0.159)
Per year of exposure, P 90th vs 10th pct.	0.006	0.007	0.010
Mean of outcome	0.728	0.728	0.728
Person-waves	39,607	39,607	39,607
Persons	6,167	6,167	6,167

Notes: Each block is a separate estimate of the first three steps of Section 4, one per column, for the outcome named, on the 1975 to 1999 cohorts described in Section 3, weighted by the PSID longitudinal weight. Exposure is the share of the 69-month window under any reform in column 1 and the strictness-adjusted share in columns 2 and 3; the coefficients are the effect of moving that share from 0 to 1. Every column carries birth-state, census-division by birth-year and survey-wave effects and the controls of Section 4. Columns 1 and 2 also carry birth-year effects, and column 3 replaces them with reform-wave by birth-year effects and is the preferred estimate. Standard errors clustered on the birth state are in parentheses and wild-cluster bootstrap p-values (999 Rademacher draws, null imposed) in brackets; stars mark significance at the 10, 5 and 1 percent levels (*, **, ***), by the bootstrap p-value for the interaction and the cluster-robust p-value for the main effect. The per-year row rescales the interaction to one year of exposure as that column measures it, for a child at the 90th percentile of P against one at the 10th, a gap of 0.453 computed over children with each counted once, that is the coefficient times 0.453 and divided by 5.75. The index is in standard deviations of the unexposed children; the components are in their own units. Outcomes are measured from age 18. The bootstrap p-value for self-rated health in column 1 is 0.098, so it carries a star at two decimals.

atric diagnosis. In column 3 each strictness-adjusted year raises the index by 0.072 standard deviations at the same contrast, a coefficient of 0.911 with a bootstrap p-value of 0.051. Both components improve, the distress scale by 0.013 per year, 1.5 percent of its mean, at a bootstrap p-value of 0.086, and the share with no psychiatric diagnosis by 1.8 percentage points per year at a p-value of 0.067. Across the columns the index rises from 0.528 to 0.891 to 0.911, so here almost the whole rise is the strictness step. The third step's reform-wave by birth-year effects change the estimate by 0.020. For a child at the 3.8 strictness-adjusted years of the fully exposed, the gain at the same contrast in P is 0.27 standard deviations.

Table 4 turns to the employment index, employment, annual hours and earnings, measured from age 24. The index does not move by an amount the estimates can detect, 0.024 standard deviations per year in column 3 with a 90 percent confidence interval from -0.014 to 0.063 and a bootstrap p-value of 0.36. The interval rules out a loss larger than 0.014 standard deviations per year and a gain larger than 0.063, so a gain up to the size of the general health gain, 0.047 per year, is not excluded, and neither is zero. The estimate for employment itself is 0.4 percentage points per year and is not significant at the 10 percent level, with a bootstrap p-value of 0.67. Hours are positive but only marginally significant, 51 more hours of work per calendar year for each year of exposure, with an interval from 13 to 90 and a bootstrap p-value of 0.10. Earnings are uninformative. The coefficient on the inverse hyperbolic sine of earnings, read as a log change, gives an interval from roughly -9 to 20 percent per year. What the hours rise means depends on what the two variables count. Annual hours count all hours worked in the year, including years with none, and employment is measured on the day of the interview, so the estimates cannot say whether the rise in hours comes from more people working part of the year or from longer hours among those already working. With the index, employment and earnings far from significance and hours at the 10 percent threshold, I read the employment index and its components as a null with a marginally significant hours estimate, and not as a finding beside the two health results. The employment estimates also rest on fewer exposed children than the health estimates, because employment is measured from age 24. The exposed children were born from 1988 on, so the youngest of them are observed at that age only in the last one or two waves. Exposed person-waves are 14 percent of the person-waves from age 24 in Table 1, against 21 percent of those from age 18. Most of these children's adult employment is still to be observed.

The fourth step of Section 4, deferred above, is Appendix Table B1. It reports the five rules entered together in the preferred specification, for the three indices. Each rule enters with its own exposure, built the way W is built but from that rule's score alone. A rule's score is 0 at the AFDC setting and 1 at the setting of that same rule that pushed a parent

Table 4: Exposure to welfare reform and adult mental health

	(1) Share under any reform	(2) Strictness-adjusted share	(3) Column 2 with reform-wave × birth-year effects
Mental health index			
Exposure $\times P$	0.528* (0.246)	0.891** (0.369)	0.911* (0.418)
	[0.08]	[0.04]	[0.05]
Exposure	0.063 (0.173)	0.089 (0.248)	-0.388 (0.471)
Per year of exposure, P 90th vs 10th pct.	0.042	0.070	0.072
Mean of outcome	-0.084	-0.084	-0.084
Person-waves	25,136	25,136	25,136
Persons	4,910	4,910	4,910
Distress scale (1 = no distress)			
Exposure $\times P$	0.099* (0.040)	0.152* (0.065)	0.160* (0.076)
	[0.05]	[0.07]	[0.09]
Exposure	-0.017 (0.028)	-0.019 (0.042)	-0.067 (0.075)
Per year of exposure, P 90th vs 10th pct.	0.008	0.012	0.013
Mean of outcome	0.818	0.818	0.818
Person-waves	23,030	23,030	23,030
Persons	4,752	4,752	4,752
No psychiatric diagnosis			
Exposure $\times P$	0.121 (0.082)	0.232* (0.118)	0.227* (0.128)
	[0.18]	[0.07]	[0.07]
Exposure	0.052 (0.049)	0.062 (0.072)	-0.126 (0.157)
Per year of exposure, P 90th vs 10th pct.	0.009	0.018	0.018
Mean of outcome	0.843	0.843	0.843
Person-waves	25,103	25,103	25,103
Persons	4,910	4,910	4,910

Notes: Each block is a separate estimate of the first three steps of Section 4, one per column, for the outcome named, on the 1975 to 1999 cohorts described in Section 3, weighted by the PSID longitudinal weight. Exposure is the share of the 69-month window under any reform in column 1 and the strictness-adjusted share in columns 2 and 3; the coefficients are the effect of moving that share from 0 to 1. Every column carries birth-state, census-division by birth-year and survey-wave effects and the controls of Section 4. Columns 1 and 2 also carry birth-year effects, and column 3 replaces them with reform-wave by birth-year effects and is the preferred estimate. Standard errors clustered on the birth state are in parentheses and wild-cluster bootstrap p-values (999 Rademacher draws, null imposed) in brackets; stars mark significance at the 10, 5 and 1 percent levels (*, **, ***), by the bootstrap p-value for the interaction and the cluster-robust p-value for the main effect. The per-year row rescales the interaction to one year of exposure as that column measures it, for a child at the 90th percentile of P against one at the 10th, a gap of 0.453 computed over children with each counted once, that is the coefficient times 0.453 and divided by 5.75. The index is in standard deviations of the unexposed children; the components are in their own units. Outcomes are measured from age 18.

Table 5: Exposure to welfare reform and adult employment

	(1) Share under any reform	(2) Strictness-adjusted share	(3) Column 2 with reform-wave × birth-year effects
Employment index			
Exposure $\times P$	0.254 (0.213)	0.373 (0.332)	0.308 (0.298)
	[0.27]	[0.33]	[0.36]
Exposure	0.148 (0.093)	0.039 (0.193)	-0.158 (0.359)
Per year of exposure, P 90th vs 10th pct.	0.020	0.029	0.024
Mean of outcome	0.127	0.127	0.127
Person-waves	29,849	29,849	29,849
Persons	5,655	5,655	5,655
Employed			
Exposure $\times P$	0.105 (0.084)	0.147 (0.136)	0.055 (0.123)
	[0.26]	[0.35]	[0.67]
Exposure	0.055 (0.041)	-0.009 (0.085)	-0.035 (0.128)
Per year of exposure, P 90th vs 10th pct.	0.008	0.012	0.004
Mean of outcome	0.801	0.801	0.801
Person-waves	29,788	29,788	29,788
Persons	5,652	5,652	5,652
Annual hours			
Exposure $\times P$	459 (247)	631 (358)	652 (297)
	[0.16]	[0.19]	[0.10]
Exposure	121 (102)	175 (191)	483 (314)
Per year of exposure, P 90th vs 10th pct.	36	50	51
Mean of outcome	1,764	1,764	1,764
Person-waves	28,350	28,350	28,350
Persons	5,120	5,120	5,120
Own labor earnings, inverse hyperbolic sine			
Exposure $\times P$	0.441 (0.776)	0.748 (1.187)	0.697 (1.100)
	[0.59]	[0.56]	[0.56]
Exposure	0.482 (0.384)	0.038 (0.717)	-1.171 (1.394)
Per year of exposure, P 90th vs 10th pct.	0.035	0.059	0.055
Mean of outcome	9.777	9.777	9.777
Person-waves	29,307	29,307	29,307
Persons	5,567	5,567	5,567

Notes: Each block is a separate estimate of the first three steps of Section 4, one per column, for the outcome named, on the 1975 to 1999 cohorts described in Section 3, weighted by the PSID longitudinal weight. Exposure is the share of the 69-month window under any reform in column 1 and the strictness-adjusted share in columns 2 and 3; the coefficients are the effect of moving that share from 0 to 1. Every column carries birth-state, census-division by birth-year and survey-wave effects and the controls of Section 4. Columns 1 and 2 also carry birth-year effects, and column 3 replaces them with reform-wave by birth-year effects and is the preferred estimate. Standard errors clustered on the birth state are in parentheses and wild-cluster bootstrap p-values (999 Rademacher draws, null imposed) in brackets; stars mark significance at the 10, 5 and 1 percent levels (*, **, ***), by the bootstrap p-value for the interaction and the cluster-robust p-value for the main effect. The per-year row rescales the interaction to one year of exposure as that column measures it, for a child at the 90th percentile of P against one at the 10th, a gap of 0.453 computed over children with each counted once, that is the coefficient times 0.453 and divided by 5.75. The index is in standard deviations of the unexposed children; the components are in their own units. Outcomes are measured from age 24. The bootstrap p-value for annual hours in column 3 is 0.103, so it carries no star at two decimals.

hardest toward work. A child's exposure to the rule is the share of her window spent under it, weighted by that score. A cell in Table B1 is therefore the effect of spending the whole window under that rule at its score-1 setting rather than under the AFDC setting, with the other four rules held fixed. For the earnings disregard the score-1 setting is the most generous one, under which a family kept part of its grant until its earnings reached the poverty line, and Appendix A gives the anchor for each of the other four. On the two health indices only one cell is marginally significant, the time limit on general health, 0.49 with a cluster-robust p-value of 0.10, and the other four rules on that index have p-values above 0.39. On mental health every p-value is above 0.28. For the employment index two rules are large and opposed, 1.27 for the disregard with a p-value below 0.01 and -1.32 for the exemption age with a p-value of 0.035, while the index itself in Table 4 is 0.31. Per year at the setting scored 1, for a child at the 90th percentile of P against one at the 10th, that is 0.10 standard deviations more employment from the disregard and 0.10 less from the exemption age.

Those two rules are on opposite sides of the choice that Section 1 put to the parent, between more money for the family and more of the parent's time at home. The earnings disregard let a working family keep more of its grant, so it raised what the family gained from the parent's work. The exemption age set how young a child could be when the parent had to work, and at the strictest setting the parent was expected to work from the month the child was born, so this rule decided when the parent's time was taken and changed nothing in what the family kept. The rule that raised what work paid raised the children's employment as adults, and the rule that took the parent's time earliest lowered it. Those are the signs that Section 1 gave the two sides of the choice. The employment index in Table 4 can be a null while two rules move employment, because its coefficient combines all five rules, and a gain from one rule and a loss from another offset in that combination. The two effects show only when each rule has its own coefficient, as it does in Table B1. The same two rules move neither health index by an amount the estimates can detect, and the exemption age is negative on general health and positive on mental health. Of the other three rules, the time limit has the largest cell on general health, but a time limit ends the grant and pushes the parent into work at the same time, so it is on both sides of the choice and does not test it. The work requirement and the sanction, the last two, are not significant on any index. So Table B1 does not settle how much of the gains in Tables 2 and 3 came from money and how much from time, because neither of the two rules moves the health indices by an amount the estimates can detect. The two rules separate money from time only in the children's employment.

6 Robustness

The preferred specification identifies β_2 from the difference in strictness across the states of a wave, so anything else that moved with that strictness across a state's cohorts would have to be kept out of it. A shock that reached all of a state's children alike goes into the main effect of exposure, β_1, and not into β_2, as Section 4 showed, so β_1 is where such a shock would show. β_1 is the effect of exposure on children whose families had no predicted chance of receipt. The rules paid those families nothing, and the check assumes that reform reached their children in no other way, such as through the local labour market, so exposure should not move their outcomes. If it does, something other than the rules moved with a state's reform across its cohorts, and β_1 is therefore a placebo for the design. Table 5 reports β_1 beside β_2, the coefficient on exposure interacted with P, for the ten outcomes of Section 5, with a cluster-robust p-value on β_1 and the bootstrap one on β_2 as in Section 4. I report both in the preferred specification and in the same regression without the census-division by birth-year effects, to show what those effects remove.

In the preferred specification β_1 is negative on nine of the ten outcomes and is not significant at the 10 percent level on any of them. On the general health index it is -0.330 with a standard error of 0.404 and a p-value of 0.41, against a β_2 of 0.600. On the mental health index it is -0.388 with a standard error of 0.471 and a p-value of 0.41, against a β_2 of 0.911. On the employment index it is -0.158 with a p-value of 0.66. Annual hours is the one outcome where β_1 is positive, 483 hours with a p-value of 0.12, against a β_2 of 652.

Without the census-division by birth-year effects, β_1 is significant at the 10 percent level on six of the ten outcomes. None of the six is in the employment family. On the two health indices it is negative and significant, -0.647 with a p-value of 0.04 on general health and -1.010 with a p-value of 0.006 on mental health. Every outcome is oriented so that a higher value is better, so a negative β_1 of that size says that children whose families had no predicted chance of receipt did worse, as their exposure rose across cohorts, in the states whose rules were stricter. That is the shape a state-cohort shock would leave if it hit all children alike. The division effects take such a shock out wherever it moved all the children of a division's cohorts together, and with them in place the placebo passes on every reported outcome. A passing placebo shows no sign of any shock of that kind.

The same two specifications also say what the effects do to β_2, the estimate of interest. On the general health index β_2 is 0.601 without the effects and 0.600 with them. On the mental health index it rises from 0.553 to 0.911, and its bootstrap p-value falls from 0.20 to 0.05, so the variation the effects remove had been pulling that estimate down. Rescaled as in Tables 2 to 4, multiplied by the 0.453 gap in P between the 90th and the 10th percentile and

divided by the 5.75 years of the window, β_2 on general health is 0.047 standard deviations per strictness-adjusted year in both specifications. The mental health estimate runs from 0.044 without the effects to 0.072 with them.

The estimates in Section 5 are valid only if nothing besides the welfare rules moved the children of likely-recipient families relative to the other children of the same birth year and reform wave, and that is the one condition β_1 cannot test. A shock that reached likely-recipient families alone, in step with the strictness of a state's rules, moves β_2 and leaves β_1 at zero. Such a shock must move with the strictness of the rules and it must reach only the families the rules reached, and few things in the setting do both. The main one that does is a state EITC adopted in the reform years. A state EITC pays cash to working families with low earnings, and the families the rules pushed into work are among them. Its effect on their children would therefore be counted as an effect of the welfare rules. Only a refundable state EITC matters here, because a nonrefundable one pays nothing to a family that owes no income tax.

By the state EITC panel of the University of Kentucky Center for Poverty Research, seven places started a refundable EITC in the reform years or after them, 1993 or later: New York in 1994, Massachusetts in 1997, Colorado in 1999, the District of Columbia in 2000, New Jersey in 2001, Oklahoma in 2002 and Rhode Island in 2003. I drop every child born in one of these seven places in or after the fifth year before its EITC started, because that child's window from conception to age five could include a year with the EITC. Those seven places keep their children born earlier, so no birth state leaves the sample. That cut removes 181 of the 6,225 children in the general health sample. On the rest, with everything else in the preferred specification kept, β_2 on the general health index rises from 0.600 to 0.668 and has a bootstrap p-value of 0.059 (Appendix Table C1). On the mental health index it moves less, from 0.911 to 0.871, with a bootstrap p-value of 0.064. On the employment index it falls from 0.308 to 0.133, with a bootstrap p-value of 0.68, so the index was a null before the cut and stays one after it. In a stricter cut I treat any rise in a state's refundable credit, measured as a share of the federal one, in the same way as a start. The rule reaches back to 1989, the first year the panel covers. Minnesota started a credit in 1991 and Vermont raised its credit in 2000, so this cut also drops their children born in or after the fifth year before that change, 220 children from the general health sample in all. Every Minnesota child in the panel was born inside that reach, so Minnesota itself leaves the sample under this cut and 50 birth states remain. The two health coefficients are then 0.672 on general health and 0.829 on mental health, with bootstrap p-values of 0.067 and 0.077. The employment index stays a null at 0.164, with a bootstrap p-value of 0.62. Under both cuts the two health estimates keep their sign and remain significant at the 10 percent level, and neither moves by more

Table 6: The placebo: exposure among children whose families had no predicted chance of receipt

	Without division effects		Preferred column	
	β_1	β_2	β_1	β_2
General health index	-0.647** (0.313)	0.601* (0.276)	-0.330 (0.404)	0.600* (0.276)
	[0.04]	[0.07]	[0.41]	[0.08]
Self-rated health	-0.521** (0.257)	0.637* (0.242)	-0.116 (0.360)	0.636* (0.252)
	[0.04]	[0.05]	[0.75]	[0.08]
Not obese	-0.227* (0.129)	0.119 (0.118)	-0.192 (0.159)	0.122 (0.112)
	[0.08]	[0.38]	[0.23]	[0.32]
Mental health index	-1.010*** (0.369)	0.553 (0.416)	-0.388 (0.471)	0.911* (0.418)
	[0.01]	[0.20]	[0.41]	[0.05]
Distress scale (higher = less distress)	-0.179*** (0.061)	0.102 (0.074)	-0.067 (0.075)	0.160* (0.076)
	[0.00]	[0.24]	[0.38]	[0.09]
No psychiatric diagnosis	-0.281** (0.113)	0.142 (0.126)	-0.126 (0.157)	0.227* (0.128)
	[0.01]	[0.29]	[0.42]	[0.07]
Employment index	-0.061 (0.236)	0.230 (0.307)	-0.158 (0.359)	0.308 (0.298)
	[0.79]	[0.47]	[0.66]	[0.36]
Employed	-0.023 (0.090)	0.047 (0.123)	-0.035 (0.128)	0.055 (0.123)
	[0.80]	[0.72]	[0.78]	[0.67]
Annual hours	13 (297)	508 (326)	483 (314)	652 (297)
	[0.96]	[0.19]	[0.12]	[0.10]
Earnings, inverse hyperbolic sine	0.301 (0.864)	0.520 (1.112)	-1.171 (1.394)	0.697 (1.100)
	[0.73]	[0.66]	[0.40]	[0.56]

Notes: The second pair of columns is one estimate of equation (2), the preferred specification, on the children of Section 3, and is column 3 of Tables 2 to 4. The first pair is the same regression without the census-division by birth-year effects. β_1 is the coefficient on exposure, the effect of strictness-adjusted exposure on a child whose family had no predicted chance of receipt ($P = 0$), and β_2 is the coefficient on exposure $\times P$, the additional effect for a certain recipient. Standard errors clustered on the birth state are in parentheses. In brackets, β_1 carries its cluster-robust p-value and β_2 its wild-cluster bootstrap p-value from 999 draws, and stars mark significance at the 10, 5 and 1 percent levels (*, **, ***) by the same p-value; the two pairs of columns come from separate runs of the bootstrap.

than 0.3 of its standard error from Tables 2 and 3. A state EITC that started or rose in the reform years therefore does not produce them. The conclusion quotes the two health figures of the preferred specification, 0.047 and 0.072 standard deviations per strictness-adjusted year, with the null β_1 behind them and both EITC cuts beside them.

7 Conclusion

This paper presents new evidence on what welfare reform did to the health and the employment of the children who were young when it reached their families. Under Aid to Families with Dependent Children a poor family with children was entitled to a cash grant with no work condition attached. Between 1993 and 1998 every state replaced that entitlement with aid that a parent had to work for, that could be cut if the parent did not, and that ran

out after a fixed number of months. As a result, a parent who complied went to work and kept the grant, so the family had more money and the child had less of the parent's time. A parent who did not comply stayed home and lost the grant, so the child had more of the parent's time and the family had less money. The early-life literature has measured what happens to children when a family's money or a parent's time rises, as it did when food stamps or Medicaid arrived (Hoynes et al., 2016; Boudreaux et al., 2016; Bailey et al., 2024). It has had less to say about a program that cut one of the two while raising the other, so the sign of the reform's effect on the children it reached was open. To estimate that effect, I need a measure of the rules each child faced. The date a state reformed is not such a measure, because the rules were the states' to write and two states that reformed in the same year could ask very different things of a parent. I therefore score each state's rules year by year on a scale from 0, the AFDC rules, to 1, the strictest setting any state used, with the work requirement, the earnings disregard, the time limit, the sanction and the work exemption weighted by the size of their employment effects in the experimental literature. I then measure a child's exposure as the months from conception to age five spent under those rules, each weighted by the score in force, so that a year under rules scored 0.5 counts as half a year. Because the rules could reach a child only through the family's welfare status, I compare children whose families were likely to be on welfare with children born in the same year whose families were not. I make that comparison only among states that reformed in the same year. Each child's chance of receipt comes from a model fitted on families observed before any state reformed, applied to her own family's characteristics over her first five years. The effect is the coefficient on exposure interacted with that chance, estimated on children born between 1975 and 1999 in the Panel Study of Income Dynamics, with birth-state, survey-wave, census-division by birth-year and reform-wave by birth-year effects, and standard errors clustered on the birth state. The same regression also estimates the effect of exposure on children whose families had no predicted chance of receipt, and that estimate is the design's placebo.

For the children of families the rules were likely to reach, each strictness-adjusted year of exposure before age five is associated with a 0.047 standard deviation higher general health index and a 0.072 standard deviation higher mental health index in adulthood. Both indices are oriented so that a higher value is better health. The scale is a child whose family stood at the ninetieth percentile of predicted receipt against one at the tenth, born in the same year in states that reformed in the same year. Both estimates are significant at the 10 percent level under a wild-cluster bootstrap over the 51 birth states. The general health gain comes mostly from self-rated health, and obesity moves the same way with less precision. Both parts of the mental health index improve, the distress scale and the share of adults with

no psychiatric diagnosis. The window from conception to age five runs 5.75 years, and the children exposed for all of it averaged 3.8 strictness-adjusted years because the rules they faced sat below the strictest setting. Over those 3.8 years, the gap between the ninetieth and the tenth percentile of predicted receipt is 0.18 standard deviations on general health and 0.27 on mental health. The gap on general health between a child whose family was certain to be on welfare and one whose family had no predicted chance of it is 0.40 standard deviations over the same years. Hoynes et al. (2016) estimate the effect of food stamps the same way, with exposure interacted with the family's chance of using the program, and for a family certain to use it they find that full access from conception to age five lowered an index of metabolic conditions by 0.4 standard deviations. The two indices are not the same measure, so the comparison is one of size only. Within this paper, the gains on both health indices are larger when a child's months under reform are weighted by the strictness of the rules in force than when those months are counted without that weight, so the estimates pick up what the rules required and not only how long they applied. The placebo holds as well once the census-division by birth-year effects are in the regression. With them, exposure moves none of the health and employment outcomes among children whose families had no predicted chance of receipt. The employment outcomes are measured from age 24, and for them I cannot reject the null. The exposed children are also a smaller share of the observations at those ages than of the health observations. Even so, two of the rules move employment in opposite directions when each is estimated with its own coefficient. Letting a working family keep more of its grant raised the children's employment as adults, and requiring a parent to work before the child turned three lowered it.

The literature on the children the reform reached has mostly found harm, and it measured that harm in childhood. Infant mortality rose as time limits began to bind (Leonard and Mas, 2008), work requirements that reached mothers of infants lowered first-year development (Herbst, 2017), preschool children's homes scored lower on emotional support (Kalil et al., 2023), and adolescents showed more delinquency and substance use (Dave et al., 2021). I argue that a harm in childhood and a gain in adulthood can both be true of the same children. Childhood is when a parent's absence and a cut grant are felt directly. The outcomes here are measured from age 18 and from age 24, by which time a parent's earnings have had years to turn into housing, food and care, so the sign of the net can differ by the age at which it is measured. Dustmann et al. (2024) find harm in adulthood as well, in a cut of about 30 percent in transfers to refugee families in Denmark that lowered the children's schooling and adult earnings. Unlike the reform studied here, that cut came with no work rule, so nothing rose as the cash fell. Dore et al. (2025) are the study closest to this one. They examine the same reform in the PSID and report the opposite sign for early exposure, 0.41 points

more on a 24-point distress scale per year of exposure among children of families likely to be eligible. The estimate here is a gain on the mental health index, and the two are in different units on different measures, so the comparison to make is the sign. Four differences in design separate the two estimates. The first is that their exposure is dated from the state's first major waiver or its start of TANF, so California's exposure starts from a 1992 waiver that raised only the earnings disregard, and California counts the same as a state that imposed full-family sanctions, whereas the strictness index here gives the disregard its own weight and scores California's other rules at zero until they changed. The second is that their design does not handle staggered adoption, so their estimate takes a state that had already reformed as a comparison for one that reformed later, whereas the estimates here compare children only with children in states that reformed in the same year. The third is that they sort children into treated and control groups on family structure and parental education recalled by the adult in the interview that records the outcome, so an adult in poor health may recall a harder childhood and be sorted into the treated group on that account, whereas the characteristics here were recorded by the parent while the childhood was happening. The fourth is that their sample also includes the children of immigrants, for whom PRWORA was an eligibility cut rather than a set of conditions, so a single reform date assigns two different treatments the same value. None of the four says on its own which estimate is right, but together they mean that the two papers measure different exposures on different children, so the two estimates do not answer the same question.

My own design does not say whether the health gains came from the family's extra money or from the parent's lost hours at home. Giving each rule its own coefficient separates money from time for employment but not for health. A family that gained money lost hours at home in the same years, so the estimate is the net of the two and not the effect of either one. Mullins (2026) separates the two in a model of maternal work and investment in children, estimated on three random-assignment welfare-to-work experiments. In that model whether a child gains depends on the family's income and on the kind of care the child gets while the parent works. My design does not observe which care arrangements these children entered. Telling money, hours and care apart would need a record of the care these children received, and that is a question for future research.

The estimate this paper reports is an intention-to-treat effect, because a child counts as more exposed when her family was likely to be on welfare, whether or not it ever received aid. I also drop the immigrant families the PSID added after 1968, because the 1996 law cut those families' eligibility instead of attaching conditions to their aid, so their children faced a different treatment from the one measured here. And the oldest exposed children reach age 35 in the data, so the chronic conditions that appear later in life are not yet observed.

These estimates matter now because work requirements are being extended to programs that did not have them and to parents who were previously excused. The 2025 reconciliation law attaches one to Medicaid (Congressional Research Service, 2025b) and widens the one in the Supplemental Nutrition Assistance Program (Congressional Research Service, 2025a). Rules that push a parent toward work take time the parent would otherwise spend with the child. The rules studied here came with a cash grant that a working family could partly keep, and under those rules the children they reached were healthier in adulthood. For the children studied here, the estimates show that the rule that let a working family keep part of its grant raised their adult employment and the rule that required a parent to work before the child turned three lowered it. How a work requirement treats earnings and young children therefore matters for the employment of the children in the families it covers.

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A Construction of the strictness index

This appendix gives what is needed to recompute the index from the policy panel. Work is the largest part of what PRWORA changed, and the index scores every rule that governs it. The statute and the waivers changed other things besides, among them the family cap, two-parent eligibility, the immigrant bar, and school and immunization requirements for a recipient's children. I leave those out because they reach a family through some channel other than the parent's job, or because no study measures what they did to employment.

The index for state s in year t is

$$S_{st} = \sum_{c=1}^{5} w_c \, I_{cst}, \tag{4}$$

where I_{cst} is the intensity of rule c in that state and year, a number between 0 and 1, and w_c is a fixed weight. The weights sum to one, so the index also runs from 0 to 1. Every intensity is 0 at the AFDC setting, so a state under AFDC on every rule scores 0. No state-year reaches 1, because no state chose the strictest setting on every rule at once. The highest value in the panel is Connecticut's 0.977 from 1996.

A.1 Scoring each rule

Two of the five rules are switches and take the value 0 or 1. The work requirement is 1 in a year in which a statewide work mandate was in force and 0 otherwise. Under AFDC the mandate was weak, since exemptions were broad and enforcement was light, so every AFDC year scores 0. The sanction is 1 when a family that did not meet the work requirement lost its whole grant and 0 when it lost only the adult's share, which was the AFDC rule.

The other three rules are quantities. Each is scored as the state's setting minus the AFDC setting, divided by the strict anchor minus the AFDC setting, and kept between 0 and 1. The time limit is measured in months of assistance a family may receive. The AFDC setting is no limit. The federal 60-month limit of 1996 is scored the same, because the federal clock did not bind for any family in the exposure window before 2001. The strict anchor is the 18-month limit Tennessee adopted, the shortest in any state. The score is

$$I_{\text{time}} = \max\left\{0, \min\left[1, \frac{60 - \text{months}}{60 - 18}\right]\right\}, \tag{5}$$

so a 24-month limit scores 0.857, 36 months scores 0.571 and 48 months scores 0.286. The exemption is the age of the youngest child, in months, below which a parent was excused

from work. Under AFDC that age was 36 months. The strict anchor is 0, immediate work, which five states adopted. The score is

$$I_{\text{exempt}} = \max\left\{0, \min\left[1, \frac{36 - \text{months}}{36}\right]\right\}, \tag{6}$$

so a 12-month exemption, the setting of 27 jurisdictions in 1998, scores 0.667. A state that set the age above 36 months is more lenient than AFDC and scores 0, since the scale does not represent leniency below the baseline.

A.2 The earnings disregard and its exit point

The disregard is scored by the exit point, the monthly earnings at which a family of three loses its grant. Under AFDC a working family kept 30 dollars plus one third of its earnings for four months and 30 dollars for the next eight. After those twelve months only the 90-dollar work-expense deduction remained. Every dollar earned above that deduction then reduced the grant by a dollar, so the sustained exit point under AFDC was

$$\text{exit}_{\text{AFDC}} = 90 + B_s, \tag{7}$$

where B_s is the state's maximum monthly grant for a family of three. A state that disregarded a flat amount F and a share p of the remainder had a grant that reached zero when the counted earnings equalled the grant. The exit point is therefore

$$\text{exit}_{\text{TANF}} = F + \frac{B_s}{1 - p}. \tag{8}$$

The strict anchor is the federal poverty line for a family of three, FPL_t, and the score is

$$I_{\text{disregard}} = \max\left\{0, \min\left[1, \frac{\text{exit}_{\text{TANF}} - \text{exit}_{\text{AFDC}}}{\text{FPL}_t - \text{exit}_{\text{AFDC}}}\right]\right\}. \tag{9}$$

Because the AFDC exit point contains the state's own grant, the same disregard formula scores differently in different states. A 400-dollar grant with a half disregard moves the exit point from 490 to 800 dollars, and a 700-dollar grant with the same formula moves it from 790 to 1,400, which is past the poverty line and scores 1. Eleven states carried the exit point to or past the poverty line and score 1. The ceiling is deliberate, since almost no family in the caseload earned above the poverty line while receiving assistance, so a disregard beyond that point changed no one's return to work. A disregard that applied only for an introductory period and then reverted to the AFDC rule scores 0, because the sustained exit point is

unchanged. Three states had such rules. The score is computed once, in the year the state adopted its own disregard rule, with that year's grant and poverty line, and held fixed until the state changed its sustained formula again. Computing it every year would have let the score drift downward as the poverty line rose against a fixed nominal grant. In this panel 121 state-years would then have registered a fall in strictness with no policy behind it.

A.3 The weights

Each weight starts from the consensus effect of the rule on parental employment, in percentage points, divided by the sum of the effects. The consensus effects were assembled from the experimental and quasi-experimental studies of each rule. Table A1 lists them with the studies behind them. For the work requirement the national estimates sit low and the experimental ones high. Fang and Keane (2004) attribute about 1.9 percentage points of the 1993 to 2002 rise in single mothers' employment to work requirements. Smedslund et al. (2006) put the pooled effect of 46 randomized welfare-to-work programs near 3 points. The three-arm Minnesota experiment isolates the mandate from the financial incentives and finds it added 6 to 10 points to employment among long-term recipients (Miller et al., 2000). Vermont's three-arm experiment found that its incentive-only arm moved employment little while the mandate arm did (Scrivener et al., 2002). The consensus of 4 points sits between the national and the experimental estimates. For the disregard the Minnesota incentive-only arm raised employment by 7 points in the first year and 3.6 later (Miller et al., 2000), while Matsudaira and Blank (2014) find that the statutory disregard expansions alone did little, because few recipients used them. The consensus of 2.5 points sits between them, and it is the least settled of the large weights. For the time limit, Grogger (2003) attributes about 0.7 points of the employment rise to time limits and Fang and Keane (2004) about 1.1, and the structural estimates of Grogger and Michalopoulos (2003) are larger, so the consensus is 1.5. For the sanction, Looney (2005) finds a clear effect on welfare use and an imprecise one on employment. Fang and Keane (2004) find that moving from lenient to strict sanction and exemption rules changes work participation by half a point, so the consensus of 1.2 points is inferred from the effect on welfare use rather than measured on employment directly. For the exemption age, Pingle (2003) finds no employment gain among mothers newly subject to work requirements and Looney (2005) about 0.7 points, so the consensus is their midpoint, 0.35.

Table A1: Consensus employment effects and the weights they imply

Rule	Consensus employment effect (pp)	Weight
Work requirement	4.0	0.419
Earnings disregard	2.5	0.262
Time limit	1.5	0.157
Sanction	1.2	0.126
Age-of-youngest-child exemption	0.35	0.037

Notes: The studies behind each consensus value are named in the text.

A.4 Onset and a worked example

A state's index is 0 in every year before its reform. Reform is dated from the first year in which a work-related rule was in force statewide, so a waiver that changed only the family cap, or that ran in a few counties, does not start a state's index. The index is free to fall within a state when a rule was loosened, since forcing it never to fall would fabricate policy. It does so in three states.

Florida in 1997, its first full year under TANF, shows the arithmetic. The work requirement was in force, so $I = 1$. The disregard was 100 percent for three months and the AFDC rule thereafter, so the sustained exit point did not move and $I = 0$. The time limit was 24 months in any 60, so $I = (60 - 24)/42 = 0.857$. The sanction removed the whole grant, so $I = 1$. The exemption ended when the youngest child was 3 months old, so $I = (36 - 3)/36 = 0.917$. The index is

$$S = 0.419 \times 1 + 0.262 \times 0 + 0.157 \times 0.857 + 0.126 \times 1 + 0.037 \times 0.917 = 0.71.$$

B The rules entered together

Table B1 reports the fourth specification of Section 4, the five rules entered together in the preferred specification. Section 5 reads it.

Table B1: The rules entered together

	General health index	Mental health index	Employment index
Work requirement $\times P$	0.269 (0.321) [0.40]	-0.320 (0.943) [0.73]	0.609 (0.495) [0.22]
Earnings disregard $\times P$	0.068 (0.213) [0.75]	0.365 (0.449) [0.42]	1.273*** (0.256) [0.00]
Time limit $\times P$	0.488* (0.294) [0.10]	0.960 (0.894) [0.28]	0.471 (0.467) [0.31]
Sanction $\times P$	0.195 (0.324) [0.55]	-0.092 (0.687) [0.89]	-0.300 (0.443) [0.50]
Exemption age $\times P$	-0.282 (0.333) [0.40]	0.660 (1.160) [0.57]	-1.315** (0.622) [0.03]

Notes: One regression per column, the specification of column 3 of Tables 2 to 4 with the strictness-adjusted exposure replaced by five exposures, one per rule, each the share of the 69-month window weighted by that rule's own 0 to 1 score, and each with its own interaction with P. The cells are the interactions. Standard errors clustered on the birth state are in parentheses and cluster-robust p-values in brackets; stars mark significance at the 10, 5 and 1 percent levels (*, **, ***). Each rule's share is on the same 0 to 1 scale as the strictness-adjusted exposure, so a cell is the effect of spending the whole window under that rule at the setting scored 1 on its scale, with the other four rules held fixed. Section 5 rescales a cell per year at the 90th against the 10th percentile of P as in Tables 2 to 4, by multiplying by 0.453 and dividing by 5.75.

C The state EITC check

Table C1: Dropping children exposed to a new or enlarged refundable state EITC

	(1) Preferred column, Tables 2 to 4	(2) Without children exposed to a refundable state EITC started in 1993 or later	(3) Column 2, also treating a rise in a credit after 1989 as a start
General health index			
Exposure $\times P$	0.600* (0.276)	0.668* (0.278)	0.672* (0.280)
	[0.08]	[0.06]	[0.07]
Exposure	-0.330 (0.404)	-0.352 (0.419)	-0.337 (0.424)
Person-waves	40,731	40,017	39,840
Persons	6,225	6,044	6,005
Mental health index			
Exposure $\times P$	0.911* (0.418)	0.871* (0.446)	0.829* (0.447)
	[0.05]	[0.06]	[0.08]
Exposure	-0.388 (0.471)	-0.197 (0.485)	-0.324 (0.484)
Person-waves	25,136	24,886	24,805
Persons	4,910	4,816	4,787
Employment index			
Exposure $\times P$	0.308 (0.298)	0.133 (0.323)	0.164 (0.316)
	[0.36]	[0.68]	[0.62]
Exposure	-0.158 (0.359)	-0.028 (0.387)	-0.003 (0.399)
Person-waves	29,849	29,537	29,444
Persons	5,655	5,521	5,491

Notes: Each column of a block is one estimate of column 3 of Tables 2 to 4 for the index named. Column 2 drops every child born in New York, Massachusetts, Colorado, the District of Columbia, New Jersey, Oklahoma or Rhode Island in or after the fifth year before the state's refundable EITC started (1994, 1997, 1999, 2000, 2001, 2002 and 2003). Column 3 treats any rise in a refundable state EITC after 1989 as a start, and so it also drops the matching children in Minnesota (1991) and Vermont (2000). State EITC rules are from the University of Kentucky Center for Poverty Research National Welfare Data. Standard errors clustered on the birth state are in parentheses. The interaction carries its wild-cluster bootstrap p-value from 999 draws in brackets; column 1 takes it from Tables 2 to 4 and columns 2 and 3 from a separate run. Stars mark significance at the 10, 5 and 1 percent levels (*, **, ***), by the bootstrap p-value for the interaction and the cluster-robust p-value for the main effect.

D Where reform happened

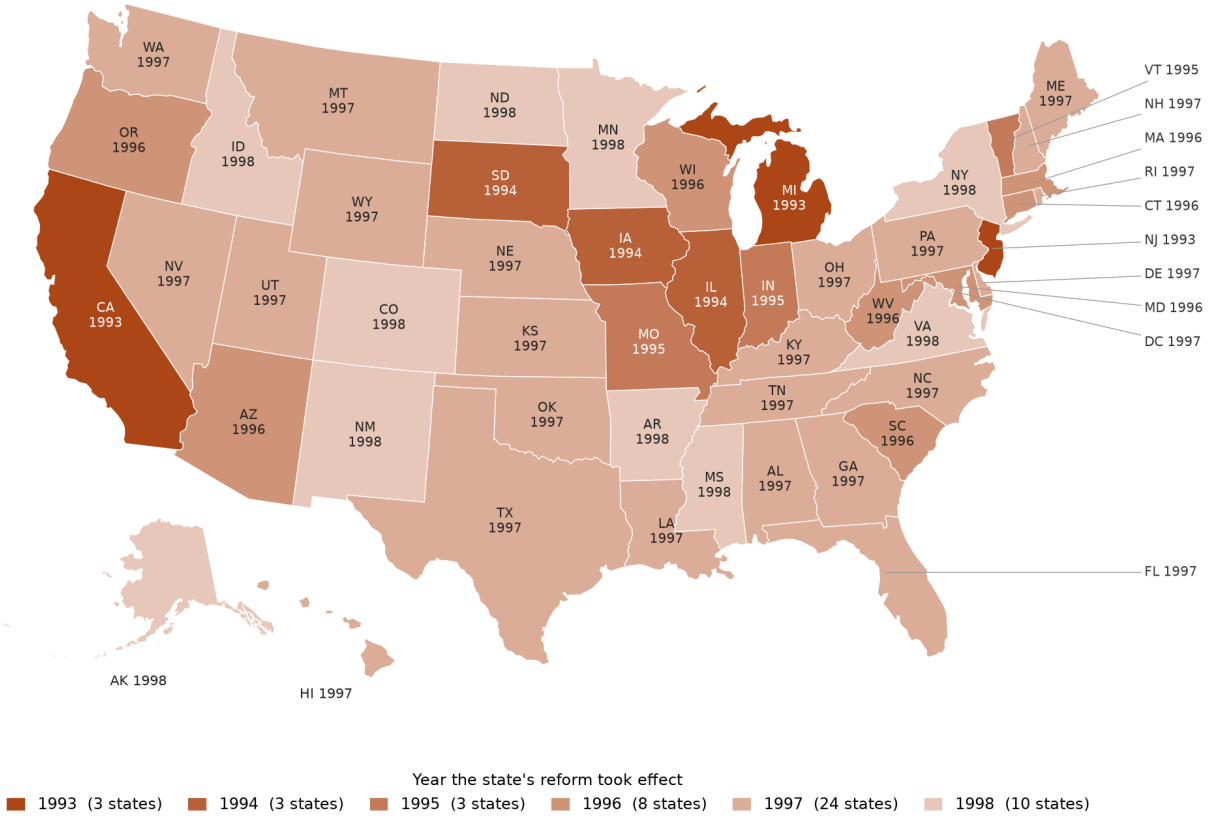


Figure D1: Year each state's reform took effect, 1993 to 1998, by the dating rule of Section 3.4. The legend gives the number of states in each year.

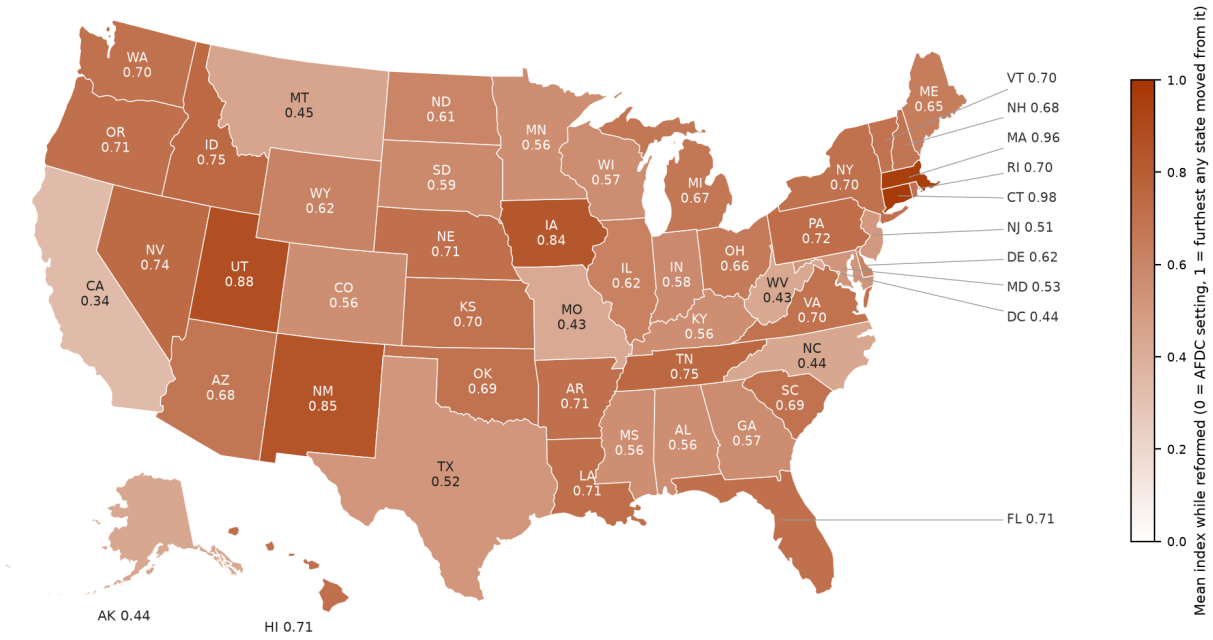


Figure D2: Mean strictness index from each state's own first year above zero through 1998. States first score above zero between 1993 and 1998, so the average covers between one and six years depending on the state. The index is 0 at the AFDC setting and 1 at the furthest any state moved away from it.